

WORKING PAPER

**Evaluating Agricultural Loans and Expenditure Allocation in Nigeria
Using Double Machine Learning**

William Hankins* Alexis Villacis[†] Ani Katchova[‡] Ivette Contreras[§]

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Abstract

Agricultural credit is widely promoted as a policy instrument for advancing food security and agricultural transformation in low- and middle-income countries. Yet evidence on how smallholders allocate the credit they receive remains mixed, particularly in contexts characterized by limited formal financial penetration. This paper examines how agricultural credit affects household expenditure allocation in Nigeria using panel data and a Double/Debiased Machine Learning (DML) framework. We estimate the causal impact of receiving an agricultural loan on farm and nonfarm expenditures and allow treatment effects to vary continuously with farm size. On average, credit increases farm and nonfarm expenditures by 14.5 percent. We also find evidence of heterogeneity: disaggregated outcome analysis identifies credit-induced fertilizer intensification that rises with farm size. The gradient is recovered by DML but absent under a conventional two-way fixed effects benchmark. Less precise aggregate results suggest total farm investment rises with landholdings, while the effect on nonfarm expenditures does not vary systematically with scale. These findings indicate that agricultural credit operates through differentiated channels depending on production capacity, supporting intensification where scale permits and primarily relaxing liquidity constraints among smaller farms. The results highlight the importance of accounting for heterogeneity in the design and evaluation of agricultural finance policies. They also show that nonparametric estimation adds the most value where expenditure responses are nonlinear in scale.

Key Words: agricultural credit, household expenditures, Nigeria, double/debiased machine learning

JEL Classification: C45, O12, O23, Q12

*The Ohio State University. Email: hankins.181@osu.edu

[†]The Ohio State University. Email: villacis.9@osu.edu

[‡]The Ohio State University. Email: katchova.1@osu.edu

[§]World Bank Development Data Group. Email: icontreras@worldbank.org

1 Introduction

Across low- and middle-income countries, agricultural credit has become a central policy instrument for advancing food security, rural development, and agricultural transformation. Governments and multilateral institutions increasingly rely on subsidized lending, credit guarantees, and risk-sharing facilities to relax liquidity constraints and promote farm investment (FAO, 2019). In principle, expanded access to agricultural finance enables farmers to invest in productivity-enhancing inputs, smooth seasonal cash flows, and manage risk.

In Nigeria, where smallholders dominate the farming population (FAO, 2018) and credit markets remain largely informal, agricultural credit occupies a prominent place in rural development policy. Programs such as the Agricultural Credit Guarantee Scheme Fund (ACGSF), the Commercial Agriculture Credit Scheme (CACS), the Nigeria Incentive-Based Risk Sharing System for Agricultural Lending (NIRSAL), and the Anchor Borrowers' Programme (ABP) aim to inject liquidity into the agricultural sector. While some initiatives have been associated with increases in aggregate production (Umeh and Adejo, 2019), credit penetration remains low: only about 5 percent of farming households report receiving loans for agricultural purposes (World Bank Group, 2025), and most loans originate from informal or semi-formal sources.

This environment raises a persistent empirical question in smallholder agricultural finance: when farmers gain access to credit, does it translate into productive farm investment, or does it primarily support consumption smoothing and risk management? Understanding how credit is deployed once received (and whether responses vary across farmers with different production endowments) remains central to evaluating credit-based interventions.

We address this question using panel data from Nigeria and a Double/Debiased Machine Learning (DML) framework. We estimate the causal impact of receiving an agricultural loan on household expenditure allocation, distinguishing between farm expenditures (inputs, land, labor, and livestock) and nonfarm expenditures (food and other household consumption). We further examine heterogeneous treatment effects as a continuous function of farm size to test whether credit responses vary systematically across the land distribution.

A large empirical literature examines whether relaxing credit constraints translates into higher agricultural investment and productivity. Early work emphasizes that credit access affects both the level and composition of farm investment under incomplete markets (Carter and Olinto, 2003). More recent experimental and quasi-experimental studies find that credit can increase agricultural investment, but that impacts are highly heterogeneous and depend on farm scale, timing, and complementary institutions such as insurance or savings mechanisms (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2020). In many settings, larger or less constrained farms are better positioned to convert liquidity into productivity-enhancing investments, while smaller farms face structural constraints that limit returns to credit-financed input use. Consistent with this, a par-

allel literature shows that agricultural and microenterprise credit frequently functions as general household liquidity rather than narrowly targeted investment finance. Borrowers often reallocate funds toward consumption smoothing, risk management, or nonfarm activities, particularly in environments characterized by income volatility, food insecurity, and missing insurance markets (Angelucci et al., 2015; Banerjee et al., 2015; Meager, 2019). These patterns align with household portfolio optimization under binding constraints, where stabilizing consumption and food security may dominate marginal on-farm investment, especially among smaller or more vulnerable farms.

Our results indicate that receiving an agricultural loan increases both farm and nonfarm expenditures by 14.5 percent on average. However, these average effects mask heterogeneity. The farm investment response appears to rise with landholdings: we identify a credit-induced 20.8 percent increase in fertilizer utilization for each additional hectare, a response paralleled imprecisely in total farm expenditures (6.9 percent per hectare, but not statistically significant). In contrast, the effect on nonfarm expenditures does not vary significantly with farm size. This is consistent with previous empirical evidence suggesting that the smallest or liquidity constrained landholders often allocate a disproportionate share of additional liquidity toward nonfarm expenditures rather than farm investment (Hussain and Thapa, 2016; Fink et al., 2020).

These findings suggest that agricultural credit operates through different channels depending on farm scale. For smaller landholders, credit primarily relaxes household liquidity constraints, while for relatively larger smallholders it supports productive intensification. From a policy perspective, the results underscore the limits of uniform credit expansions and highlight the importance of scale-sensitive credit design and complementary risk-mitigation mechanisms.

Our paper contributes to the literature on agricultural credit and smallholder behavior in three ways. First, we provide new evidence on the allocation of agricultural credit across uses in Nigeria by documenting how loan access affects both farm and nonfarm expenditures. Rather than focusing solely on production outcomes, our expenditure-based approach captures multiple channels through which credit relaxes household constraints, including consumption smoothing, risk management, and farm investment. Second, we contribute to the literature on heterogeneous treatment effects by estimating how expenditure responses vary continuously with farm size, rather than grouping farms into broad size categories. We document evidence consistent with a systematic relationship in which smaller landholders allocate a larger share of liquidity toward nonfarm expenditures, while farmers with greater land endowments increasingly channel credit toward productive farm investment, helping reconcile mixed evidence on the investment impacts of agricultural credit. Third, benchmarking panel-adjusted Double Machine Learning against analogous two-way fixed-effects specifications, we show that the value of nonparametric estimation is context dependent: the two estimators converge on nonfarm routine expenditures but diverge on the farm investment margin, where the household response is nonlinear in scale. This analysis

demonstrates how flexible estimation can aid the recovery of policy-relevant, plausibly causal estimates in environments characterized by high-dimensional covariates and non-random program participation. In contexts such as Nigeria, where experimental variation is limited and credit programs are implemented at scale, our approach offers a practical framework for evaluating food system interventions while accounting for household heterogeneity.

The remainder of the paper is organized as follows. Section 2 describes the institutional context and data. Section 3 outlines the empirical strategy. Section 4 presents results on average and heterogeneous effects of agricultural credit on expenditure allocation. Section 5 discusses the findings and their limitations, and Section 6 concludes and draws policy implications.

2 Study Setting

2.1 Institutional Context

Although Nigeria has long been characterized as Africa’s largest oil exporter, oil revenues have declined as a share of GDP over the past three decades. Over the same period, agriculture has become increasingly central to the domestic economy, surpassing oil in its contribution to GDP and accounting for roughly one-quarter of total employment. Despite this economic importance, the agricultural sector remains highly fragmented and dominated by smallholders, with an average farm size of approximately 0.5 hectares. Development challenges are substantial. Survey evidence suggests that nearly 70 percent of smallholder farmers live below the poverty line, most operate at subsistence levels where a large share of output is consumed rather than marketed, and only about 7 percent report access to formal credit (FAO, 2018). In theory, relaxing liquidity constraints in such settings should generate high marginal returns to agricultural investment (Ali et al., 2014; Carter and Olinto, 2003). In practice, however, binding consumption needs and income volatility may lead households to allocate additional liquidity toward food, essential goods, or risk management rather than exclusively toward farm intensification (Karlan et al., 2014; Suri et al., 2021; Bocher et al., 2017). Low credit penetration therefore presents both a constraint and a policy opportunity, and credit expansion has the potential to improve household welfare while strengthening domestic food production.

Nigeria’s farm credit market is segmented across formal, public, and informal institutions. Formal providers include commercial banks, microfinance institutions, and public entities such as the Nigeria Agricultural Cooperative and Rural Development Bank (NACRDB). Informal and semi-formal mechanisms—including family networks, cooperatives, and rotating savings associations (e.g., Ajo or Esusu)—play a critical role in filling gaps left by the formal sector (Adetiloye, 2012). Nonetheless, overall credit access remains limited. Formal lenders face high transaction costs, collateral constraints, and information asymmetries that restrict lending to smallholders, while

informal lenders lack sufficient capital to meet broader demand (Awotide et al., 2015; Osabohien et al., 2020; Ozoko et al., 2022; Ukwuaba et al., 2020). This segmentation reflects broader patterns documented in the credit literature: marginal enterprises operating at very small scales often prioritize immediate consumption and liquidity needs when credit becomes available, whereas larger producers are better positioned to translate borrowing into productive investment (Banerjee et al., 2015; Hussain and Thapa, 2016).

In response to these constraints, the Federal Government and the Central Bank of Nigeria have repeatedly implemented directed lending and risk-sharing programs. The Agricultural Credit Guarantee Scheme Fund (ACGSF), established in 1977, guarantees up to 75 percent of loan defaults for participating lenders. Subsequent reforms introduced mechanisms such as the Self-Help Group Linkage Banking model, the Trust Fund Model, and the Interest Drawback Scheme, all intended to reduce effective borrowing costs and encourage lending to smallholders. Other initiatives, including the Family Economic Advancement Programme, sought to expand and partially formalize traditional savings mechanisms (FAO, 2013; Obinne, 1999). Despite these efforts, credit to agriculture remained limited, averaging roughly 3 percent of total national credit between 1978 and 2006 (Adetiloye, 2012).

More recently, the Anchor Borrowers' Programme (ABP), launched in 2015, aimed to address coordination failures through an interlinked transaction model connecting banks, farmers, and off-takers. The program provided credit guarantees while requiring farmers to sell output to designated buyers to reduce default risk. At its peak, the ABP facilitated over one trillion Nigerian naira (approximately USD 2.7 billion, c. 2020) in loan commitments and has been associated with increases in domestic grain production (Umeh and Adejo, 2019). However, repayment rates were reported to be approximately 25 percent (IMF, 2023), and amid tightening fiscal conditions, new loan applications were suspended in 2024.

These developments occurred alongside substantial macroeconomic shocks. Border closures in 2019 contributed to rising food prices, followed by oil price volatility in 2020 that constrained government revenues. Subsequent fiscal adjustments included the unwinding of programs such as the ABP, the removal of fuel subsidies, and exchange rate reforms. Together, these shocks have intensified food insecurity¹ and heightened pressure on rural households. In this environment, understanding how credit is allocated once accessed is particularly salient. Distinguishing whether loans support agricultural intensification or primarily facilitate consumption smoothing is central to evaluating the effectiveness of credit-based food system interventions.

¹For the interested reader, we present evidence of the rising food insecurity among Nigerian households over the period 2018-2024 in Figure A1 of the supplementary Appendix A.

2.2 Living Standards Measurement Study Core Data

We use data from the Nigeria General Household Survey – Panel (GHS-Panel), part of the World Bank’s Living Standards Measurement Study–Integrated Surveys on Agriculture (LSMS-ISA). The GHS-Panel is nationally representative and designed for policy analysis. Our analysis draws on the two most recent waves (Wave 4: 2018–2019; Wave 5: 2023–2024), each covering approximately 5,000 households, with roughly 89 percent successfully re-interviewed across waves. Each wave includes household, community, and agricultural questionnaires collected during post-planting (July–September) and post-harvest (January–March) phases (NBS, 2019, 2024).

The household module records demographic characteristics, asset ownership, income sources, economic shocks, and food consumption. We aggregate individual-level information to the household level (the unit of analysis), using head-of-household characteristics as demographic controls. The agricultural module is recorded at the plot level and includes land tenure, plot size, crop production, and fertilizer use. Plot-level variables are aggregated to the household level for key measures such as total cultivated area and fertilizer use.

We construct two primary outcome variables: farm expenditures and nonfarm expenditures. Farm expenditures include land (rent or purchase), livestock (purchase and upkeep), crop inputs (seeds, fertilizer, machinery, pesticides), and hired labor. Nonfarm expenditures include imputed food consumption and non-food expenditures (durables, non-durables, and services), excluding nonfarm business investments, education, health, vehicles, ceremonies, debt, rent, and holidays to isolate regular household consumption. All expenditure variables are annualized to ensure comparability across recall periods.²

Key control variables include farm land area, value of crop production, durable assets, livestock holdings, nonfarm income, land rights, economic shock indicators, food security measures, and demographic characteristics.

To construct the analytical sample, we harmonize variable definitions across waves and standardize categorical variables where classifications differ. We restrict our analysis to households engaged in agriculture (2,410 observations dropped by design) and that can be paired across waves (883 observations dropped by attrition), cumulatively reducing the sample from 9,691 to 6,398 observations, corresponding to 3,199 agricultural households observed in both waves. For specifications involving farm-size interactions, we restrict to 5,900 observations with non-missing land area. As discussed in Section 3, nuisance functions are trained using the full paired panel with imputation to maximize predictive performance, while final treatment effect estimates are reported on the restricted subsample to maintain consistency across models. Extreme values in assets, expenditures, and land area are winsorized at the 99th percentile.

Table A2 defines the key variables used in the analysis, Table 1 reports summary statistics for

²Food expenditures are based on 7-day recall, while some agricultural expenditures use longer recall windows. All are converted to annual values.

the full sample by wave, and Tables A3, A4, and A5 report quartile summary statistics (pooled and by survey wave). The proportion of landholders receiving agricultural loans is uniformly low across survey waves, with approximately 4.9 and 4.8 percent of households in wave 4 and 5, respectively (full sample N=6,398). Nominal loan values increased between waves 4 and 5, with the mean loan size increasing from ₦76.7 to ₦129.8 thousand. Notably, the second pooled quartile (Table A3) reports the highest mean loan size (₦110.2 thousand), indicating that loan amounts do not increase monotonically with farm size. This is noteworthy given the relatively large differences in landholdings, with the average farm size rising from 0.11 hectares in the bottom quartile to 3.16 hectares in the top quartile (an increase of more than 20-fold across quartiles).³

Expenditure patterns differ substantially across the farm size distribution. Households in the top quartile have considerably higher farm expenditures but lower nonfarm expenditures than those in the bottom quartile. Although expenditure profiles vary sharply with land endowments, access to credit remains limited across the distribution, consistent with broad credit rationing.

2.3 Loans, Economic Shocks, and Food Security Data

The LSMS household questionnaire conducted during the post-planting visit records loan applications in the twelve months preceding the survey, approvals, sources, and intended use.⁴ Pooling data across waves, roughly one-third of loan applications were earmarked for farming, with approximately 76 percent of agricultural loans sourced from informal or semi-formal sources (e.g. friends, money-lenders, savings associations).⁵

To approximate the effect of agricultural credit programs (such as the Anchor Borrowers Program), we use loans specifically earmarked for farm expenditures as a proxy for agricultural credit.⁶ We pool loans across informal and formal sources for two reasons: (i) both sources are effectively fungible in relaxing the household liquidity constraint, and (ii) the scarcity of formal agricultural loans necessitates pooling to preserve statistical power.

We define credit access at the extensive margin (a binary treatment variable), because relaxing binding budget constraints often induces structural changes in household expenditure behavior, such as ‘crowd-in’ effects (Banerjee et al., 2019). Furthermore, this approach mitigates endogeneity concerns regarding the correlation between loan magnitude and household assets. Additionally, to isolate the treatment effect and account for households with multiple loans, we include a control dummy variable for non-agricultural loans. We choose to exclude non-agricultural loans from the

³More details about the distribution of farm size by loan access can be found in Figure A2 of the supplementary Appendix A.

⁴The specific wording of the survey questionnaire is "In the last 12 months, have you or anyone else in the household attempted to borrow money or applied for or received a loan from [a list of plausible sources]?" There is no indication that loans were made “in-kind” and specific values in Naira are provided for each loan received.

⁵See Figure A3 in Appendix A for more details on the distribution of farm size by loan formality.

⁶The LSMS datasets do not include information on whether a loan originates from a specific government program, or if it otherwise could be formally defined as an agricultural loan.

TABLE 1.
Summary Statistics by Wave

	Sample Mean			Min	Max
	Wave 4	Wave 5	Pooled		
Agricultural Loan Variables					
Received Agricultural Loan (Binary 0/1)	0.05	0.05	0.05	0.00	1.00
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	76.72	129.83	102.85	2.00	1,500.00
Outcome Variables					
Input Expenditures (1,000s ₦) ^a	26.40	40.01	33.20	0.00	501.50
Land Expenditures (1,000s ₦) ^a	4.57	4.02	4.29	0.00	215.00
Labor Expenditures (1,000s ₦) ^a	66.87	62.27	64.56	0.00	1,052.25
Livestock Expenditures (1,000s ₦) ^a	19.61	47.79	32.31	0.00	796.00
Farm Expenditures (1,000s ₦)	106.99	128.98	117.98	0.00	1,385.15
Food Expenditures (1,000s ₦) ^a	1,338.82	1,737.89	1,538.36	0.00	66,861.29
Non-Food Expenditures (1,000s ₦) ^a	287.87	567.46	427.67	3.24	9,163.29
Nonfarm Expenditures (1,000s ₦)	1,626.70	2,305.36	1,966.03	13.12	67,818.62
All Fertilizer Use (kg/ha)	0.87	0.63	0.75	0.00	47.24
Ratios					
Agricultural Loan to Total Farm Expenditures	71.7%	100.7%	87.2%		
Agricultural Loan to Nonfarm Expenditures	4.7%	5.6%	5.2%		
Control Variables					
Farming Land Area (Hectares) ^{a,b}	1.40	0.97	1.18	0.01	8.00
Value of Farm Production (1,000s ₦) ^{a,b}	338.65	372.75	355.75	0.00	3,606.00
Assets (1,000s ₦) ^{a,b}	193.05	360.31	276.68	0.00	7,735.00
Nonfarm Income (1,000s ₦) ^{a,b}	272.69	427.05	349.87	0.00	5,520.00
Livestock Holdings (#) ^{a,b}	0.93	0.65	0.79	0.00	11.90
Has Formal Land Rights (binary 0/1) ^{a,b}	0.17	0.12	0.14	0.00	1.00
Income Shock (#)	0.25	0.15	0.20	0.00	5.00
Production Shock (#)	0.38	1.52	0.95	0.00	8.00
Price Shock (#)	0.19	1.94	1.07	0.00	5.00
Head Marital Status (binary 0/1) ^c	0.78	0.76	0.77	0.00	1.00
Head Age ^c	50.02	53.47	51.74	10.00	130.00
Head Sex (binary 1 ≡ Male) ^c	0.83	0.79	0.81	0.00	1.00
Number of Members in the Household	5.91	5.94	5.93	1.00	29.00
Number of Adult Members in the Household	2.88	2.98	2.93	0.00	16.00
Household Has Phone Access (binary 0/1)	0.94	0.92	0.93	0.00	1.00
Household Has Internet Access (binary 0/1)	0.36	0.37	0.36	0.00	1.00
Probability of Moderate Food Insecurity	0.33	0.58	0.46	0.03	1.00
Food Consumption Score	51.43	46.61	49.02	0.00	112.00
Received Non-Agricultural Loan (binary 0/1)	0.10	0.07	0.09	0.00	1.00

^a Winsorized to top 1%.

^b Imputation performed by Evans School Policy Analysis and Research Group ([UW EPAR, 2025](#)).

^c Characteristics of the Head of Household, coded verbatim from survey responses.

treatment variable to more closely proxy the effects of formal agricultural loans rather than general household liquidity relaxation.

To control for exogenous disturbances, we use the Economic Shocks module within the household questionnaire during the post-harvest visit, which records unexpected events affecting the

household’s welfare in the three years preceding the survey. Because the specific shock categories vary slightly between waves (21 categories in Wave 4 and 28 in Wave 5), we harmonize them into four distinct categories: (1) Production Shocks (e.g. drought, floods, pests), (2) Income Shocks (e.g. death of earner, loss of remittances), (3) Price Shocks (e.g. fuel price inflation, input/output price volatility), (4) Other Shocks (e.g. conflict, theft).⁷ We restrict the shock window to the two years preceding the survey wave to reflect acute events which might induce or be ameliorated by credit acquisition. Specifically, we sum the number of shocks experienced by the household in each category to capture the acute intensity of shocks experienced by the household not absorbed by fixed effects.

Finally, we control for food security status. We measure household food security using the Food Consumption Score and the Food Insecurity Experience Scale (FIES), calculated using data from the Food Security module present in the household questionnaire post-harvest visit in both waves. For FIES, we follow [Cafiero et al. \(2018\)](#) and estimate the probability of a household experiencing either moderate or severe food insecurity.

3 Empirical Framework

3.1 Estimation Strategy

Our objective is to estimate how access to agricultural credit affects household expenditure allocation, and whether these effects vary with farm size. We exploit the panel structure of the LSMS data, using within-household variation between Wave 4 (2018–2019) and Wave 5 (2023–2024) to control for time-invariant unobserved heterogeneity.

A conventional two-way fixed effects (TWFE) model would address household-level fixed characteristics and common time shocks. However, given the high dimensionality of observable covariates and the potentially complex selection into credit, we adopt a Double/Debiased Machine Learning (DML) framework⁸. DML operates by partialling out the confounding effects of observables from both outcome and treatment variables, leveraging high-dimensional machine learning functions to isolate pseudo-random variation in credit access⁹. This approach preserves the fixed-effects structure while allowing flexible, non-parametric modeling of the relationships between covariates, treatment assignment, and outcomes. In our results section, and in supplementary tables in Appendix A, we benchmark our panel-adjusted DML against analogous TWFE specifications.

⁷The Other Shocks category is not included in the analysis due to its extremely sparse prevalence and no detectable effect on nuisance function performance.

⁸For our analysis, we primarily utilize the DoubleML package developed for Double Machine Learning in Python ([Bach et al., 2022](#)).

⁹DML necessarily relies upon the assumption of conditional independence; as such, we explore the robustness of our results to omitted variable bias in Section 4.5.

In the following subsections, we outline the DML specification, its adaptation to panel data, and the implementation details of our estimation procedure.

3.2 Identification

3.2.1 Double/Debiased Machine Learning (DML) Specification

To isolate the causal effect of agricultural loan use on household expenditures while accounting for the high-dimensional confounders pervasive in credit acquisition, we employ a Double/Debiased Machine Learning (DML) approach. Unlike the linear TWFE benchmark, DML allows for non-parametric modeling of the relationship between covariates, treatment assignment, and outcomes, which can improve identification when compared to traditional econometric techniques (Belloni et al., 2014; Chernozhukov et al., 2018, 2025; Semenova and Chernozhukov, 2020). This flexibility is useful in our setting given the largely informal and opaque credit markets accessible to small-holders.

We estimate a Partial Linear Regression (PLR) model with an interaction term, defined as:

$$Y_{it} = \theta_1 L_{it} + \theta_2 (L_{it} \times \hat{A}_{it}) + g(X_{it}, \bar{X}_i) + u_{it} \quad (1)$$

where Y_{it} is the outcome and denotes each category of expenditure for household i at time t (Expenditures), L_{it} is the treatment and denotes whether or not the household received an agricultural loan (Received Agricultural Loan), and $\hat{A}_{it}$ represents the centered farm size in hectares (Farming Land Area) defined as:

$$\hat{A}_{it} = A_{it} - \bar{A}, \quad (2)$$

where $\bar{A}$ is the sample mean across all households i and survey waves t . In equation 1, the unobserved, non-parametric function $g(\cdot)$ captures confounding variance for each category of expenditure separately through both time-varying controls X_{it} and their Mundlak means $\bar{X}_i$, calculated for each household over time, controlling for time-invariant heterogeneity. We first estimate the model without the interaction term to obtain the baseline treatment effect, and then include the interaction term to test for heterogeneity based on farm size. Following Chernozhukov et al. (2025), we treat the interaction term $(L_{it} \times \hat{A}_{it})$ as a second conditional average treatment effect (CATE) such that $\hat{A}_{it} \in X_{it}$. This allows us to jointly estimate the baseline average treatment effect θ_1 , and the heterogeneous treatment effect θ_2 . The equations are estimated separately for each category of farm or nonfarm expenditures, and then for each of the sub-categories in farm expenditures (input, land, labor, livestock expenditures) and nonfarm expenditures (food and non-food expenditures).

As a linear benchmark, we also estimate analogous TWFE specifications of the form:

$$Y_{it} = \alpha_i + \lambda_t + \theta_1 L_{it} + \theta_2 (L_{it} \times \hat{A}_{it}) + \gamma X_{it} + u_{it} \quad (3)$$

where α_i are household fixed effects, λ_t are the survey wave fixed effects, and X_{it} contains the same control covariates used in the DML specification. These estimates are used as a benchmark to contextualize whether our DML results substantively differ from linear panel specifications.

3.2.2 DML in Panel Data

Adapting DML to a panel dataset requires careful handling of unobserved heterogeneity. Traditional DML assumes independence of observations, which is violated by repeated household observations as is the case here. Additionally, producing TWFE inference from DML is challenging and demeaning data prior to processing can introduce bias. To address these challenges, we implement two modifications aligned with the best practices emerging from the newly expanding DML panel data literature (Fuhr and Papies, 2024; Clarke and Polselli, 2026).

First, we incorporate Correlated Random Effects (CRE) by augmenting the control matrix X_{it} with household-level time means (Mundlak means) of all time-varying controls. Specifically, for each time-varying control (x_{it}), we include

$$\bar{X}_i = \frac{x_{i1} + x_{i2}}{2}.$$

The nuisance function is therefore written as $g(X_{it}, \bar{X}_i)$, where $\bar{X}_i$ denotes the vector of household-level means. Including these Mundlak means allows the DML specification to account for time-invariant unobserved heterogeneity, emulating a two-way fixed effects specification which would otherwise be biased in the DML framework (Clarke and Polselli, 2026). Additionally, we include a time dummy for the second time period (second wave of the survey) inside X_{it} to account for aggregate shocks. Taken together, these modifications allow DML to produce results analogous to a traditional TWFE model, but with the non-parametric flexibility of DML.

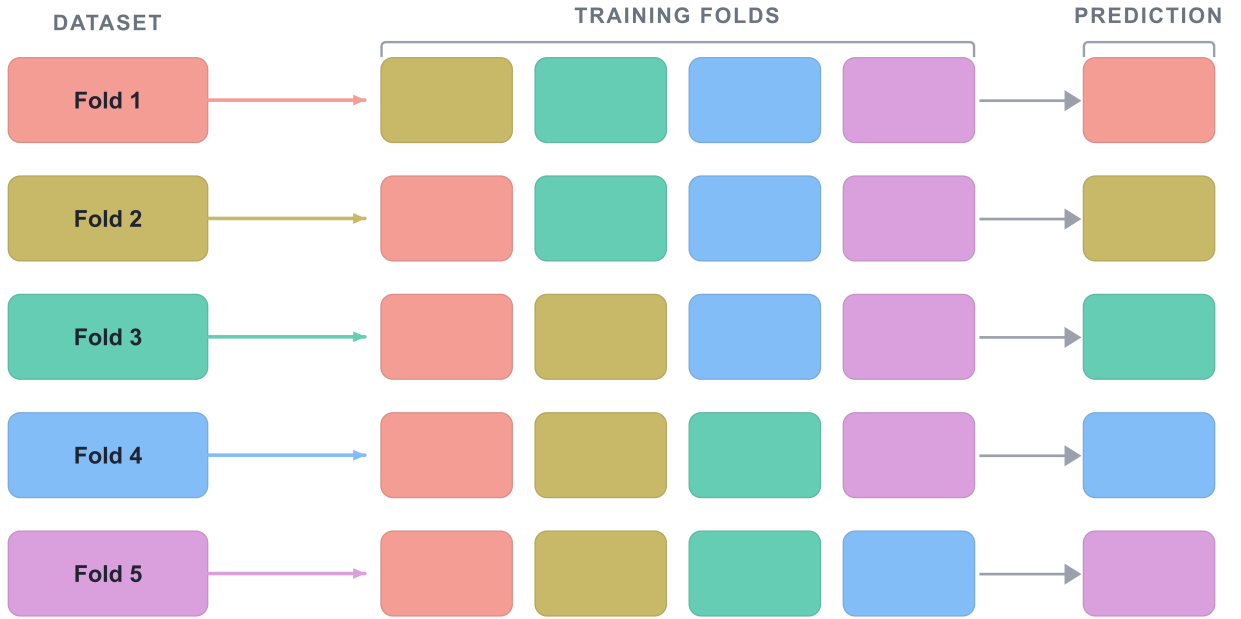
Second, we enforce grouped cross-fitting. Cross-fitting, illustrated in Figure 1, is a standard technique for avoiding overfitting. The dataset is first divided into subsets ("folds"); a model is trained on one group of folds and predictions are made on the held-out fold. For example, in Figure 1, row 1, a supervised machine learning (ML) model is trained on Folds 2 through 5 (k_2, k_3, k_4, k_5) and used to predict Fold 1 (k_1). The goal is to avoid the bias created when ML models fit idiosyncratic noise in the training data, referred to as overfitting.

Specifically, the cross-fitting procedure relies on the independence of clusters when dividing data into folds to ensure idiosyncratic noise is uncorrelated between folds. Independence between folds implies overfitting bias from the training folds is orthogonal to the prediction folds. To ad-

dress within-household inter-temporal correlation we group data at the household level when creating folds. Formally, given a household with observations i_1 and i_2 in waves 1 and 2 respectively, we require

$$i_1 \in k_l \iff i_2 \in k_l$$

for each fold k_l . So, both observations for the same household are always assigned to the same fold. This grouped cross-fitting procedure maintains independence across household clusters during nuisance function estimation.



Notes: In each row, nuisance models are trained on four folds and used to predict the held-out fold for both the outcome variable and treatment propensity. Folds are grouped at the household level, with both waves of a household assigned to the same fold.

FIGURE 1

3.2.3 Stacked Learners and Rich Controls

In the DML framework, the conditional expectation of the outcomes $\hat{E}[Y|X]$ and the treatment $\hat{E}[L|X]$ serve as nuisance function parameters labeled $\hat{n}(\cdot)$ and $\hat{m}(\cdot)$ respectively — that is, functions that must be estimated to partial out confounding but are not the core objective of the analysis. Formally, DML is a two-stage estimation process, first nuisance functions are estimated via cross-fitting and partialled out

$$\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i) \quad \text{and} \quad \tilde{L}_{it} = L_{it} - \hat{m}(X_{it}, \bar{X}_i).$$

Second, treatment effects are estimated on the partialled out outcome and treatment variables

$$\tilde{Y}_{it} = \theta_1 \tilde{L}_{it} + \theta_2 (\tilde{L}_{it} \times \hat{A}_{it}) + u_{it}.$$

Therefore, the performance of the DML estimator hinges on the predictive accuracy of our machine learning models used to approximate these functions. Consequently, to maximize model performance, we implement a specific training and estimation strategy.

First, regarding sample composition: due to a logarithmic transformation on outcomes Y , we exclude household observations with zero values for their expenditures from our models, resulting in smaller sample sizes for sub-category analysis. Additionally, to maintain consistency across model specifications and maximize first-stage accuracy, we estimate $\hat{E}[L|X]$ ($\hat{m}(\cdot)$) once on the full paired panel ($N = 6,398$ pooled observations) and reuse it across outcome-specific models. We employ median imputation with missingness indicators to leverage the complete information set. However, the outcome nuisance function ($\hat{n}(\cdot)$) and final second-stage estimation of treatment effects are restricted to the paired non-missing farm size and outcome variable subsample ($N \leq 5,900$ pooled observations) to ensure a common estimation sample for baseline and heterogeneous specifications. This strategy allows our nuisance function to benefit from maximum data depth while ensuring consistent nuisance function estimation across specifications. Specifically, outcome nuisance functions ($\hat{n}(\cdot)$) are implemented on each outcome specific sample. Our strategy of training the treatment nuisance function on the maximal available sample prior to subgroup evaluation directly follows the Group ATE framework in [Chernozhukov et al. \(2025\)](#).

Second, we employ a "stacked" ensemble for our learners by training several different types of models and a rich set of covariates in X_{it} . Ensemble learners are nested ML functions composed of a meta-learner and stacked learners, for example

$$\hat{m}(\cdot) = \hat{f}(\hat{r}(\cdot), \hat{t}(\cdot), \hat{w}(\cdot))$$

where $\hat{f}(\cdot)$ is a meta-learner such as ridge, and $\hat{r}(\cdot)$, $\hat{t}(\cdot)$, $\hat{w}(\cdot)$ are stacked learners such as forest, gradient boosting, or regularized models.

We use ensemble learners as a best practice technique for nuisance function estimation to improve the stability and generalization of our results. Our ensemble integrates a diverse set of algorithms to balance bias and variance: (i) Tree Based Models, including Random Forest and Histogram Gradient Boosting to capture non-parametric relationships, and (ii) Regularized Linear Models, including Lasso, Ridge, and Elastic Net to handle high-dimensionality and overfitting.

We use distinct learners for our outcome and treatment variables. For the binary agricultural loan treatment, we use classification variants of the learners (e.g. Logistic Regression with Elastic Net penalty, and stacking classifier), while we employ their regression counterparts for the continuous expenditures outcome. The meta-learners for each ensemble are a Ridge model for the

continuous outcome and a Logit for the binary treatment. Finally, to compute the interaction term, the residual agricultural loan treatment variable was interacted with the centered farm size ($\hat{A}$).

Complementing this robust learner set, X_{it} comprises a rich vector of socio-economic and fixed effect variables. As we mentioned above, to account for time-invariant household heterogeneity, we calculate Mundlak means for all time-varying controls and cluster at the household level. Additionally, we include state and time fixed effects, corresponding to the Nigerian Federal states and the wave indicator variables, respectively. Finally, we address covariates' missingness¹⁰ in the nuisance stage by using median imputation with missingness indicators, consistent with the missingness indicator approach in Chernozhukov et al. (2025). To avoid leakage, median imputation is fit within each training fold.

3.2.4 Inference and Robustness

To ensure the stability of our results, we repeat the cross-fitting procedure over 30 randomized repetitions and aggregate results using the median (Chernozhukov et al., 2018, 2025). We assess predictive validity of the nuisance functions using three primary diagnostics. For the outcome model, we report a standard R^2 (R_Y^2) value. For the treatment propensity model, we report the Area Under the Curve (AUC) and Average Precision (AP). We select these complementary metrics to address the specific challenges of our data: AUC¹¹ provides a standard measure of discrimination often used in traditional econometrics, while AP¹² offers a superior benchmark for unbalanced treatment and is commonly used in machine learning model assessment. Ideally, these metrics indicate that the learners are identifying genuine signals without overfitting to noise. To ensure our estimates are robust to fold specification, we test the robustness of our baseline 5-fold estimates against alternative 3- and 8-fold specifications. Finally, as a robustness check, we explore the sensitivity of our results to different cross-fitting specifications in Section 4.4 and perform a pseudo omitted variable analysis in Section 4.5.

4 Results

This section presents the empirical results from the DML models benchmarked against analogous TWFE specifications. We interpret the estimates as reflecting changes in household expenditure

¹⁰612 of 5,900 paired observations (10.4%) have at least one missing value among the control variables in our dataset, excluding observations with missing farm size values.

¹¹The Area Under the Curve measures the probability a model correctly ranks a randomly chosen treated observation higher than a randomly chosen untreated observation. A value of 0.5 indicates predictive power equivalent to random choice, while a value greater than 0.5 indicates the model assigns treatment probabilities more accurately than random chance.

¹²Average Precision summarizes a model's precision recall curve over all decision thresholds ($p(\text{treatment}|X) \geq \tau, \tau \in [0,1]$). A naive model that assigns treatment at random will yield an AP equal to the sample prevalence of treatment. Therefore, AP is beneficial to contextualize the model's predictive power in settings such as ours where treatment is unbalanced and rare.

allocation after receiving agricultural credit. Because credit relaxes liquidity constraints at the household level, observed expenditure responses capture reallocation across farm and nonfarm uses rather than adherence to stated loan purposes. We first report baseline average treatment effects, then examine heterogeneity by farm size, followed by disaggregated expenditure analyses and robustness checks.¹³ Note that calculations in this section may deviate from direct analysis using table output due to rounding.

4.1 Baseline results

Table 2 reports the baseline average treatment effects (ATE) of receiving an agricultural loan on household expenditures. Because the outcomes are log-transformed and treatment is binary, coefficients are interpreted as percentage changes using $100 \times (e^{\beta} - 1)$. Observations with zero expenditures are excluded, so estimates reflect intensive-margin responses for households with positive expenditures.

Our DML results indicate that receiving an agricultural loan increases both farm and nonfarm expenditures by 14.5 percent, statistically significant at the 10 percent and 1 percent levels, respectively. These estimates exploit within-household variation across survey waves and control for observed covariates, household correlated random effects, and common time shocks.

These results are consistent with experimental evidence showing that relaxing an agricultural household’s budget constraint raises both agricultural investment and nonfarm spending. For example, [Delavallade and Godlonton \(2023\)](#) show that warrantage (inventory credit) in Burkina Faso increases total per-capita expenditures by 13 percent (driven by non-food categories) and raises crop revenues by roughly 15 percent. Similarly, [Beaman et al. \(2023\)](#) find sizable expenditure responses to liquidity provision in Mali.

By comparison, the TWFE estimates indicate no statistically significant effect on farm expenditures and a 16.8 percent increase in nonfarm expenditures, statistically significant at the 5 percent level. These results are informative as a linear benchmark for our primary DML specification. For nonfarm expenditures, the TWFE and DML point estimates are closely aligned, suggesting that the positive consumption response is not heavily driven by the nonparametric relationships captured by DML. This is similarly consistent with the expectation that routine household consumption—recall nonfarm expenditures represents daily consumption excluding lumpy consumption (Table A2)—is more mechanically related to observable household needs and liquidity constraints. That is, we posit nonfarm expenditures are driven by observable characteristics (e.g. household size, food requirements, and cash on hand) in a way a linear specification can capture, not a literal passthrough of loan proceeds (which we discuss in Section 5).

By contrast, farm expenditure responses may depend on a more complex and interdependent

¹³A balance table and additional descriptive figures are reported in Appendix A Table A1.

Table 2.
Average Treatment Effects of Agricultural Loans on Expenditures

	Farm		Nonfarm	
	Expenditures (log)		Expenditures (log)	
	DML	TWFE	DML	TWFE
Received Agricultural Loan	0.136*	0.072	0.135***	0.155**
	(0.071)	(0.095)	(0.049)	(0.070)
<i>Specification</i>				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	No	Yes	No
Household FEs	No	Yes	No	Yes
<i>Learners</i>				
Loan $\sim X$	PLH-L		PLH-L	
Expenditures $\sim X$	ERH-R		ERH-R	
K-folds	5		5	
<i>Diagnostics</i>				
R_Y^2 (Outcome: $Y \sim X$)	0.351		0.420	
AUC (Loan $\sim X$)	0.626		0.630	
AP (Loan $\sim X$)	0.083		0.082	
Adjusted- R^2	0.352		0.429	
Within Adjusted- R^2	0.074		0.089	
Observations	5,476	5,476	5,900	5,900

Standard errors in parentheses. Columns 1 and 3 are Double Machine Learning (PLR), $n_{\text{rep}} = 30$; columns 2 and 4 are TWFE with cluster-robust standard errors at the household level.

Adj- R^2 and Within Adj- R^2 report TWFE model fit.

R_Y^2 reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

set of factors, such as farm scale, land quality, or prior success. These relationships are difficult to model parametrically and may be nonlinear or interactive in a way that a TWFE specification cannot fully capture. The difference between the TWFE and DML estimates for farm expenditures therefore suggests that the flexible nuisance function partialling out procedure in DML is most valuable for the farm expenditure margin, where the expenditure response is likely to be less mechanical. We interpret the close alignment on nonfarm as indicating that the nonparametric specification has limited additional value where the response is mechanical, but explore its potential value on the farm margin in the heterogeneity and disaggregation results below where

the relevant evidence lies.

This characterization is also supported by the relative magnitudes of variance captured in outcomes by the nuisance function (R_Y^2) and the TWFE specification (Adjusted- R^2)¹⁴. Both the DML and TWFE specifications capture significantly more variance in nonfarm expenditures than farm expenditures (0.420 vs 0.351 and 0.429 vs 0.352, respectively). These statistics highlight three important insights: first, our adjustments to DML for panel data effectively capture comparable levels of variance to a TWFE specification when substituting CREs for household fixed-effects; second, our learner ensemble and a linear fixed-effects model closely agree that nonfarm expenditures can be more effectively predicted than farm expenditures with the available controls; and third, despite controlling for comparable variance in the partialling out and prediction steps for both outcomes, point estimates only differ materially for farm expenditure coefficients. This further refines our speculation that nonfarm routine expenditures see limited benefit from nuisance function flexibility, while nonparametric nuisance function estimation may be particularly advantageous for farm expenditures.

The predictive validity of our nuisance functions is substantiated by the Area Under the Curve (AUC) and Average Precision (AP) statistics reported in Table 2. The AUC is a common diagnostic statistic for binary outcome models, and can generally be interpreted as the probability that a random positive instance is ranked higher than a negative one. Our AUC values of 0.626 and 0.630 (farm and nonfarm expenditure models, respectively) outperform the naive (random) assignment model benchmark of 0.5. While these scores suggest relatively poor discrimination by traditional standards (Hosmer et al., 2013), they align with recent literature on the difficulty of predicting credit behavior in opaque and noisy smallholder microdata. For instance, Björkegren and Grissen (2020) demonstrate that machine learning models constrained to traditional demographic data often yield AUC scores between 0.51 and 0.57 for credit repayment outcomes.

The AP is a standard machine learning metric that summarizes the precision-recall curve. It can generally be interpreted as the average probability that a positive prediction made by the model is correct, benchmarked against the baseline probability of random selection (sample prevalence). Our AP values of 0.083 and 0.082 (farm and nonfarm models, respectively) indicate moderate discrimination above random assignment. Given the low prevalence of agricultural credit in our sample (4.9 percent), a naive model would achieve an AP of only 0.049. Thus our models achieve a precision roughly 67.7–71.0 percent higher than a naive benchmark. Collectively, these diagnostics indicate that despite the opaque nature of informal credit markets where soft variables (e.g. trust, reputation) heavily influence credit access, our learners successfully isolate systemic selection signals necessary for DML nuisance function estimation.

¹⁴Note R_Y^2 and Adjusted- R^2 serve similar but distinct purposes. The R_Y^2 statistic characterizes nuisance function fit, not incorporating the treatment effect, while Adjusted- R^2 captures the overall TWFE specification fit, incorporating the treatment effect.

4.2 Heterogeneous treatment effects by farm size

We next examine whether the impact of agricultural credit varies with farm size. Table 3 augments the baseline specification with an interaction between loan receipt and farm land area (centered at the sample mean), providing suggestive evidence that farm expenditure responses increase with farm size. To supplement our analysis, we also conduct a DML quartile-based group average treatment effect (GATE) analysis, shown in Appendix B.

Table 3.
Heterogeneous Effects of Agricultural Loans on Expenditures

	Farm		Nonfarm	
	Expenditures (log)		Expenditures (log)	
	DML	TWFE	DML	TWFE
Received Agricultural Loan (Loan)	0.124* (0.072)	0.044 (0.097)	0.139*** (0.049)	0.160** (0.070)
Loan $\times$ Farming Land Area (HA)	0.067 (0.045)	0.142** (0.066)	-0.031 (0.031)	-0.029 (0.041)
<i>Specification</i>				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	No	Yes	No
Household FEs	No	Yes	No	Yes
<i>Learners</i>				
Loan $\sim X$	PLH-L		PLH-L	
Expenditures $\sim X$	ERH-R		ERH-R	
K-folds	5		5	
<i>Diagnostics</i>				
R^2_Y (Outcome: $Y \sim X$)	0.351		0.420	
AUC (Loan $\sim X$)	0.626		0.630	
AP (Loan $\sim X$)	0.083		0.082	
Adjusted- R^2		0.353		0.429
Within Adjusted- R^2		0.075		0.089
Observations	5,476	5,476	5,900	5,900

Standard errors in parentheses. Columns 1 and 3 are Double Machine Learning (PLR), $n_{\text{rep}} = 30$; columns 2 and 4 are TWFE with cluster-robust standard errors at the household level.

Adj- R^2 and Within Adj- R^2 report TWFE model fit.

R^2_Y reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The baseline coefficient on loan receipt remains positive and statistically significant for farm expenditures (0.124 log points, approximately 13.2 percent). The coefficient on the interaction term is also positive (0.067) but not statistically significant at traditional levels (P-value=0.142), suggesting that the farm expenditure response increases by 6.9 percentage points for each additional hectare of land, although it lacks the statistical precision to be distinguished from zero. In contrast, the interaction term for nonfarm expenditures is negative, notably smaller in magnitude (-0.031), and statistically insignificant (P-value=0.313), suggesting that the estimated 14.9 percent increase (0.139 log points) in nonfarm expenditures does not vary systematically with farm size.

The TWFE estimates provide a useful linear benchmark for these heterogeneous effects. Mirroring the baseline estimates in Table 2, the nonfarm expenditure DML and TWFE estimates are closely aligned: both specifications indicate a large significant ATE (0.139 and 0.160, respectively) and a small negative heterogeneous treatment effect (HTE) (-0.031 and -0.029, respectively). This supports the interpretation that agricultural credit relaxes household liquidity constraints for routine expenditures in a way that is broadly similar across the farm size distribution.

For farm expenditures the comparison is more nuanced, and estimators diverge in both level and slope: the ATE is positive and significant under DML (0.124, P-value=0.085) but small and insignificant under TWFE (0.044), while the linear interaction is significant under TWFE (0.142, P-value=0.032) but imprecise under DML (0.067, P-value=0.142). Having benchmarked the baseline DML estimates against TWFE, we now consider the source of their divergence on the farm margin. The two specifications differ in how they remove the main effect of scale: TWFE conditions on farm size linearly, whereas DML's nuisance function conditions on it nonparametrically. Because farm expenditure increases nonlinearly with scale (rising steeply among the smallest farmers before flattening as illustrated in Figure A4), and likely depends on scale jointly with other covariates, a linear control may struggle to orthogonalize scale from farm expenditures, leaving residual curvature that a nonparametric nuisance function removes.

Figures 2 and 3 illustrate the efficacy of both model specifications in orthogonalizing the outcome with respect to farm size. Each top panel plots the residuals of the two outcome models against farm size: for DML the cross-fitted nuisance residual ($\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i)$), and for TWFE, the residual from an analogous specification estimated without the treatment terms, $\tilde{Y}_{it} = Y_{it} - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})$, so that both residual series retain treatment variation and differ only in how covariates and fixed effects are partialled out. If a specification adequately captures the dependence of the outcome on farm size, then the residuals should have no curvature with respect to it. Curvature that remains admits two interpretations: either the specification has failed to fully absorb scale dependent variation in the outcome, whether directly or jointly with other covariates (functional form misspecification), or curvature is generated by variables excluded from the specification by construction (i.e. the treatment itself). For nonfarm expenditures (Figure 2),

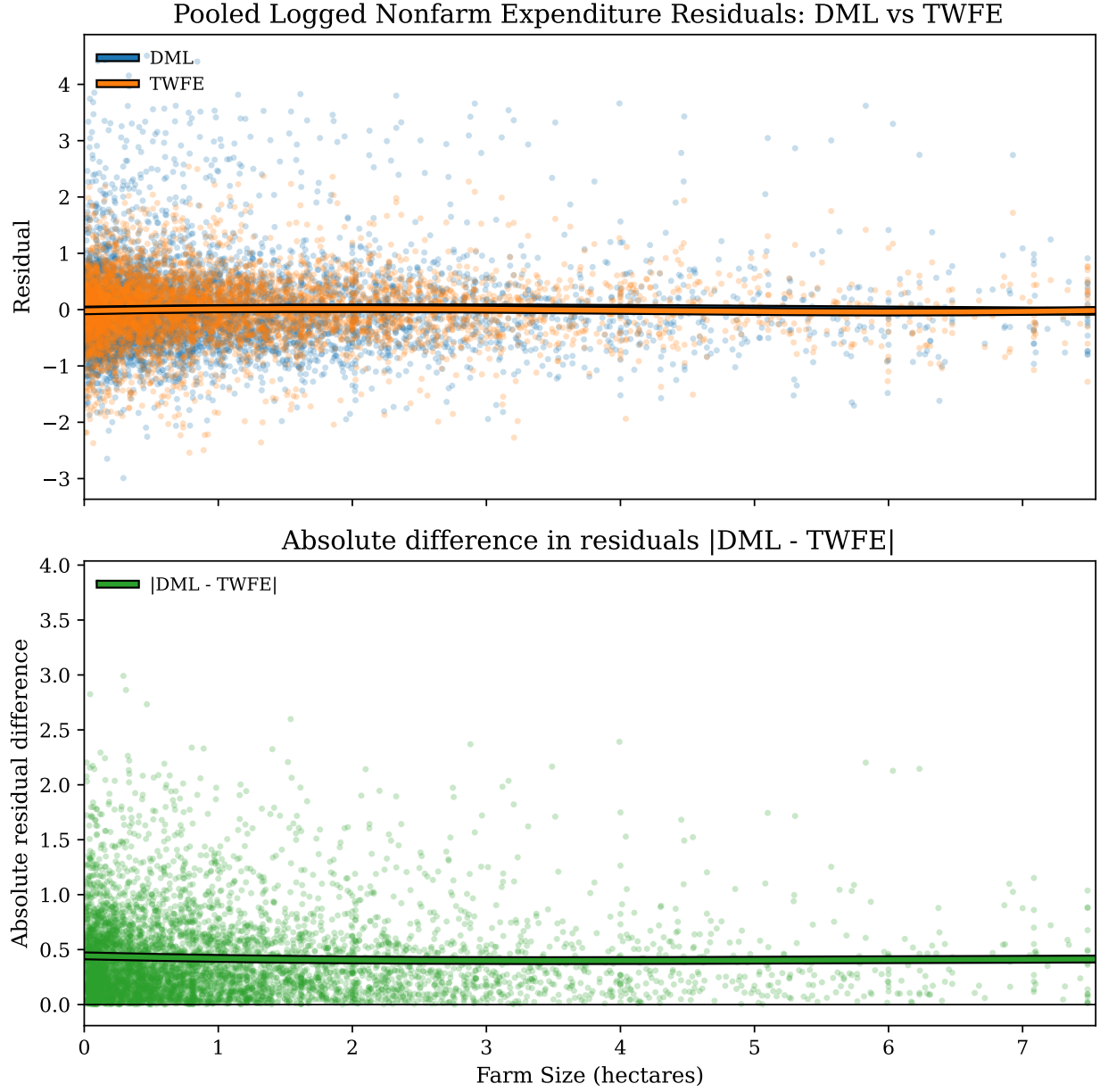
both specifications yield distributions with best fit curves¹⁵ that nearly perfectly align, implying both specifications effectively orthogonalize the outcome to farm scale. For farm expenditures (Figure 3), the TWFE best fit curve bends with farm size while the DML curve stays nearly flat. As a summary measure, we report the standard deviation (SD) of each fitted curve evaluated over the support, where a perfectly flat (orthogonalized) curve has an SD of zero. The SD of the DML fitted curve in Figure 3 is less than half of the TWFE specification (0.05 and 0.13, respectively). We can compare this with the near identical SD of the best fit curves in Figure 2, which are both approximately 0.02. If anything, this comparison handicaps DML since TWFE can benefit from in-sample overfitting while DML’s grouped cross-fitting restricts predictions to out-of-fold observations. The flatness of the DML curve is therefore unlikely to be an artifact of an overfit learner.

The difference in residuals is not just a matter of aggregate distribution but also one of allocation. The two outcome models explain comparable aggregate variance (R_Y^2 and $\text{Adj-}R^2$ are nearly identical in Table 3), so the divergence in residual curvature reflects how each specification explains variance across the scale dimension. The lower panels show the observation-level disagreement between model specifications ($|\tilde{Y}_{it}^{DML} - \tilde{Y}_{it}^{TWFE}| = |\hat{n}(X_{it}, \bar{X}_i) - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})|$). The difference in residual values illustrates the gap between the two models’ fitted values: the mean value is roughly 50 percent larger, with a 36 percent larger SD, for farm expenditures than non-farm expenditures ($\mu = 0.638$ and $\sigma = 0.535$ versus $\mu = 0.424$ and $\sigma = 0.394$, respectively). This further supports the intuition that the DML specification largely aligns with the TWFE specification in estimating routine nonfarm expenditures, but yields novel insight in the farm expenditure setting. Functional form therefore matters materially more on the farm margin.

This is also why the aggregate interaction trades places between the model specifications, through two channels. First, the lower panels of Figures 2 and 3 show that the observation level disagreement between the two specifications is roughly 50 percent larger, and more dispersed on the farm margin than on the nonfarm margin. The two models are therefore genuinely fitting different expectation functions for farm expenditures, and the second stage treatment effect estimation inherits that difference. Second, the residual curvature in the upper panels identifies how the difference manifests in our point estimates: because the interaction regressor is a function of farm size, scale-correlated residual distribution that the linear control fails to absorb projects directly onto the TWFE interaction coefficient, whereas the DML nuisance function removes it before the interaction is estimated.

That said, residual curvature is not, in itself, evidence of misspecification. Both residual series deliberately retain treatment variation, so curvature generated by credit effects that vary with farm size—the heterogeneity we are investigating—is expected. Critically, however, this treatment-driven component is common to both residual series, which are constructed from the same observations, control set (X), and treatment variation. The differential in curvature (the SD of the DML

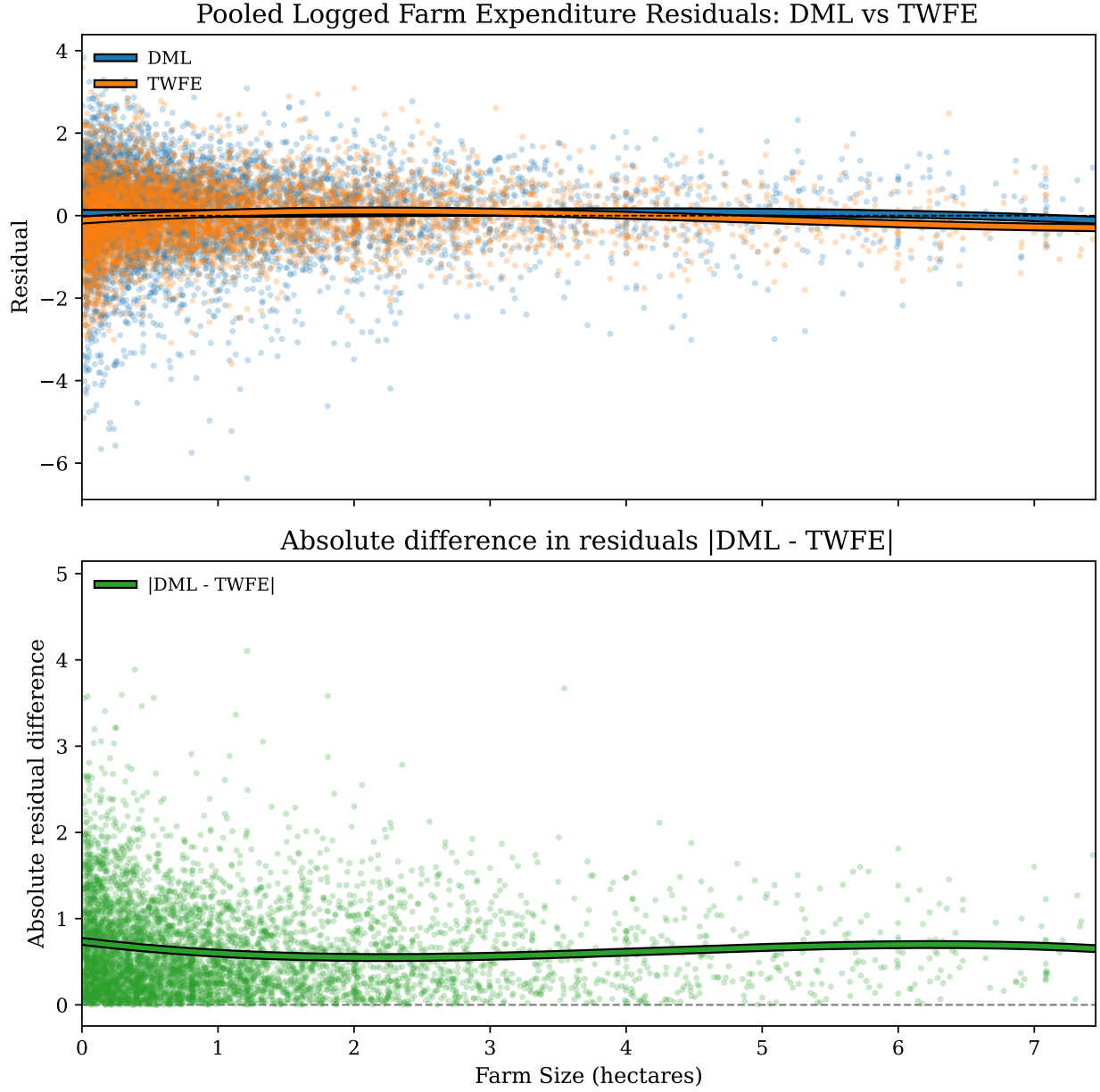
¹⁵All best fit curves in Figures 2, 3, and 5 are third-degree fitted polynomials.



Notes: Top panel: DML residuals are out-of-fold nuisance residuals, $\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i)$; TWFE residuals are from an analogous specification estimated without the treatment terms, $\tilde{Y}_{it} = Y_{it} - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})$. Solid curves are third-degree polynomial fits. Bottom panel: absolute observation-level difference between the two residual series.

FIGURE 2

curve is less than half that of TWFE, despite relying on out-of-fold predictions) therefore cannot be attributed to treatment effects, and instead indicates that the linear specification absorbs scale dependent variation less effectively. Accordingly, while we cannot determine how much of the remaining DML curvature reflects true credit effects, the comparison is consistent with a TWFE interaction coefficient that partially absorbs unmodeled scale curvature rather than treatment het-



Notes: Top panel: DML residuals are out-of-fold nuisance residuals, $\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i)$; TWFE residuals are from an analogous specification estimated without the treatment terms, $\tilde{Y}_{it} = Y_{it} - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})$. Solid curves are third-degree polynomial fits. Bottom panel: absolute observation-level difference between the two residual series.

FIGURE 3

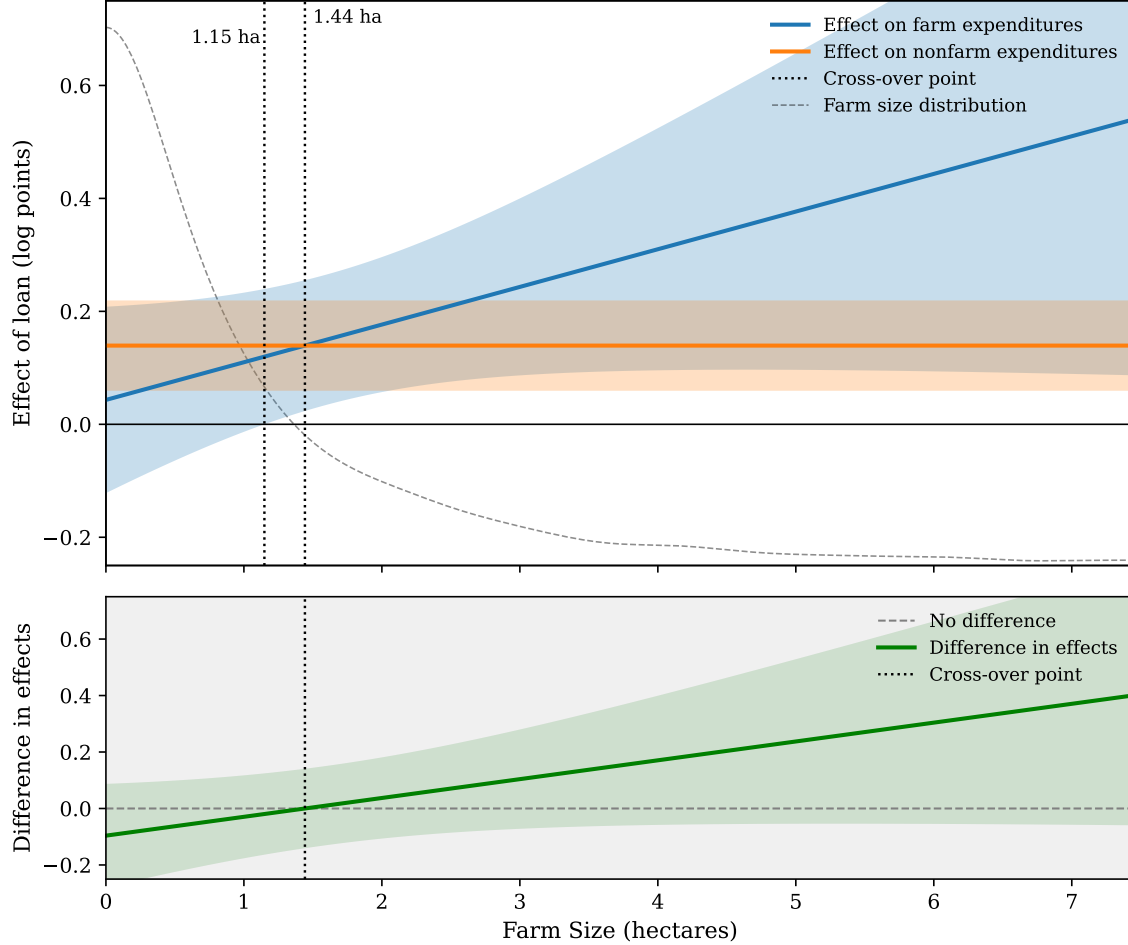
erogeneity alone. We therefore treat the DML interaction estimate as imprecise rather than as a definitive null.

To identify the heterogeneous pattern implied by the DML specification on farm expenditures, we calculate the farm size at which the estimated farm expenditure response equals the nonfarm response. Because the interaction term is centered at the mean farm size across waves (1.22 ha),

the marginal effect of agricultural credit on farm expenditures is given by the baseline coefficient plus the interaction effect scaled by the deviation from the mean. We solve for the farm size tipping point (A^*), where the increasing farm investment effect intersects the constant nonfarm consumption effect:

$$\underbrace{0.139}_{\text{Nonfarm Effect}} = \underbrace{0.124 + 0.067 \times (A^* - 1.22)}_{\text{Farm Effect}}$$

Conditional Loan Effects by Farm Size and the Johnson–Neyman Cross-Over - 90% Bands



Notes: Marginal effects of agricultural credit by farm size, from the DML estimates in Table 3, with 90 percent confidence bands and the Johnson-Neyman cross-over. The dashed curve shows the farm size distribution.

FIGURE 4

Solving for A^* yields a threshold of approximately 1.44 hectares where we estimate the relative response to farm expenditures exceeds that of nonfarm expenditures. This corresponds to approximately the 72nd percentile of the farm size distribution¹⁶. As demonstrated in Figure 4, following Brambor et al. (2006) and utilizing the Johnson-Neyman technique for statistical differ-

¹⁶This threshold is calculated using unrounded point estimates and the mean farm size specific to the farm expenditure subsample ($N = 5,476$).

ence between coefficients, we cannot confidently identify the tipping point at conventional levels (90 percent bands shown). That said, Figure 4 is informative in illustrating that the combined effect of agricultural credit on farm expenditures only becomes statistically distinct from zero when landholdings exceed 1.15 hectares. By comparison, if we take the nonfarm expenditure response as stable—or decreasing as implied by the imprecise nonfarm interaction term—across landholdings, then we observe the converse pattern: agricultural credit has an equal or larger effect on nonfarm expenditures among smaller landholders. As such, the threshold of 1.44 hectares should not be interpreted as a precisely estimated structural breakpoint, but as an imprecise descriptive summary: below this threshold, agricultural credit appears to induce a greater marginal effect on nonfarm expenditures; above it, the marginal effect of credit on farm investment dominates.

Taken together, the DML and TWFE estimates suggest that receiving an agricultural loan increases both routine nonfarm expenditures and farm investment. The nonfarm expenditure response is robust across estimators and appears broadly scale invariant. The farm expenditure response is more sensitive to the estimator, but both specifications indicate that larger farms may be better positioned to leverage scale to convert credit into productive farm spending. This pattern is consistent with the interpretation that landholdings and credit access may act as complements, though the estimated effects should be interpreted cautiously given the imprecision of the DML interaction term. Consistent with this, we treat the threshold as descriptive and locate the substantive heterogeneity evidence in the input-channel analysis below.

Importantly, these heterogeneous effects are unlikely to be driven by differences in loan magnitude. As shown in Table A3, the ratio of agricultural loans to farm expenditures is higher among the smallest farms (170.6 percent in Quartile 1) than among the largest farms (57.6 percent in Quartile 4). Despite receiving loans that are large relative to their existing farm expenditures, smaller farms exhibit weaker farm investment responses. This pattern suggests that the observed heterogeneity reflects differences in expenditure allocation rather than differences in credit volume.

4.3 Sub-category analysis for farm and nonfarm expenditures

To further examine the specific mechanisms driving the effect of agricultural loans on household expenditures we perform analysis based on the components of farm expenditures (Inputs, Land, Labor, and Livestock) and nonfarm expenditures (Food, and Nonfood). We provide additional results from a TWFE specification in Appendix A, Tables A6 and A7.

Table 4 presents the disaggregated farm expenditure treatment effects. For input, land, and labor expenditures there are no statistically significant results. This is likely a result of heterogeneity in investment strategies across households (i.e. some households intensify through input expenditures such as fertilizer utilization, while others hire labor or acquire land) and reduced power due to smaller sample sizes. However, coefficient signs are universally positive, and receiving

agricultural loans increases livestock expenditures by 44.2 percent (0.366 log points), significant at the 10 percent level. Given that this category includes recurring costs such as feed and healthcare, this pattern is consistent with credit supporting ongoing capital maintenance rather than extensive margin herd expansion¹⁷. These results are consistent in sign with the TWFE specification in Table A6, although the DML specification notably differs in magnitudes. However, the null result on disaggregated input expenditures (Column 1) may obscure specific treatment effects, so we turn from input spending to a physical measure of input use, fertilizer application intensity, observed on the positive use subsample (N=1,676, against 3,670 for input expenditures).

Table 4.
Heterogeneous Effects of Agricultural Loans on Disaggregated Farm Expenditures

	Input Exp. (log) (1)	Land Exp. (log) (2)	Labor Exp. (log) (3)	Livestock Exp. (log) (4)	Fertilizer (log kg/ha) (5)
Received Agricultural Loan (Loan)	0.045 (0.098)	0.094 (0.220)	0.054 (0.082)	0.366* (0.205)	0.010 (0.114)
Loan $\times$ Farming land area (HA)	0.066 (0.062)	0.037 (0.184)	0.037 (0.049)	0.032 (0.088)	0.189*** (0.070)
<i>Specification</i>					
Controls	Yes	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes	Yes	Yes
<i>Learners</i>					
Loan $\sim$ X	PLH-L	PLH-L	PLH-L	PLH-L	PLH-L
Expenditures $\sim$ X	ERH-R	ERH-R	ERH-R	ERH-R	ERH-R
K-folds	5	5	5	5	5
<i>Diagnostics</i>					
R_Y^2 (Outcome: $Y \sim X$)	0.362	0.234	0.238	0.226	0.386
AUC (Loan $\sim$ X)	0.605	0.661	0.620	0.633	0.599
AP (Loan $\sim$ X)	0.079	0.117	0.089	0.097	0.085
Observations	3,670	352	4,010	1,176	1,676

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R_Y^2 reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

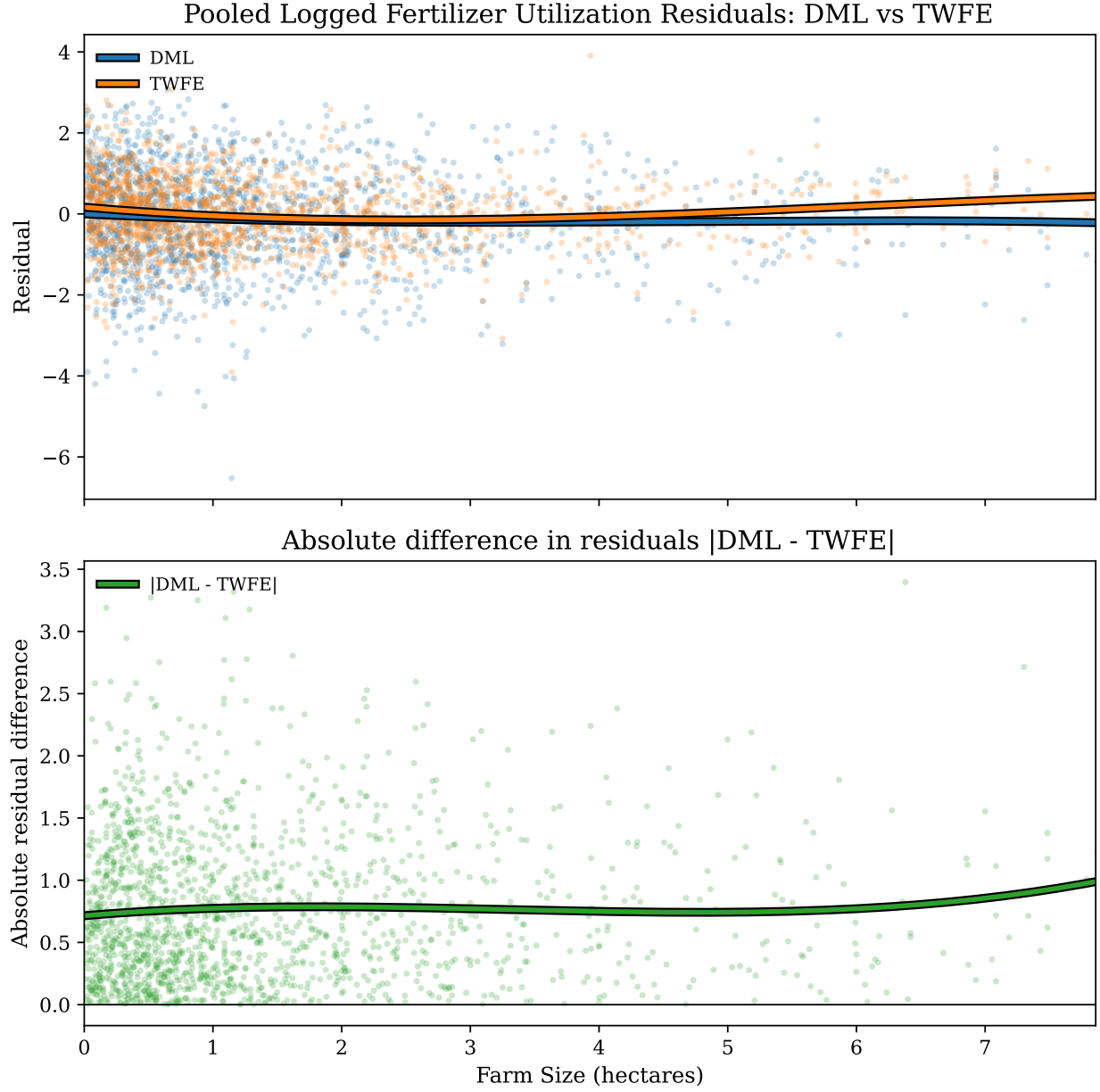
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

¹⁷The one component where the linear benchmark is the more precise of the two is land (Table A6: 0.440, significant at the 10 percent level). We do not interpret this divergence in either direction, as this specification is by a large margin the least powered analysis with $N = 352$.

To test whether the size-dependent investment effects observed in our heterogeneous effects model (Table 3) are driven by agricultural intensification, we specifically examine fertilizer application intensity (kg/ha) in Column 5. The results suggest that fertilizer intensification is important for larger smallholders who receive credit. The coefficient (0.189 log points) on the interaction term is positive and statistically significant at the 1 percent level, indicating that each additional hectare increases the credit effect on fertilizer intensity by 20.8 percentage points. Of particular note, this positive estimate runs opposite to the descriptive gradient we observe in Table A3, where the smallest farms utilize fertilizer more intensively (Quartile 1: 0.86 kg/ha) than the largest farms (Quartile 4: 0.39 kg/ha), easing concern that the gradient simply reflects the association between farm size and fertilizer use. Similarly, Table A6 differs in its point estimates, indicating a large insignificant ATE and a small insignificant HTE. This pattern is consistent with a nonlinear fertilizer investment decision that the TWFE specification does not recover.

We again interpret the two specifications against one another through their residual distributions (Figure 5), applying the diagnostics developed in Section 4.2. The same pattern observed for total farm expenditures is even more pronounced for fertilizer utilization. The fertilizer fitted curve is close to flat under DML but visibly curved under TWFE, and the gap between model specifications is even larger: the SD of the fitted curve across the support is 0.04 for DML versus 0.18 for TWFE, against 0.05 and 0.13 for total farm expenditures. This places the DML residual near the benchmark observed where both specifications agree (0.02 on the nonfarm margin), while the TWFE value is the largest in our analysis. As in Section 4.2, the loan's contribution to residual curvature is identical across the two series, so the widened differential cannot reflect credit effects. Instead, the gap indicates the linear specification absorbs scale dependent variation in fertilizer use even less completely than in aggregate farm spending. The lower panel corroborates the magnitude of the model disagreement: the observation-level gap between model residuals (equivalent to the gap in predictions) is larger for fertilizer than for aggregate farm expenditures ($\mu = 0.764$ and $\sigma = 0.595$ versus $\mu = 0.638$ and $\sigma = 0.535$, respectively).

Taken together, two features make this our central evidence of scale dependent investment. First, the average effect is essentially zero (0.010, P-value=0.933) while the interaction is strong, so the result is not that credit raises fertilizer use on average but that it does so only as scale rises. Second, the gradient is recovered by DML but absent under TWFE (see Table A6: -0.006, P-value=0.937). This is consistent with our residual analysis above (Figure 5), where the DML nonparametric estimation more thoroughly removes farm size driven variation in fertilizer utilization, leaving the interaction to be estimated against a residual purged of the scale curvature that contaminates its linear counterpart. Moreover, it is unlikely that an overfit learner would reproduce the theoretically expected sign, at high significance, precisely at the relevant margin. Indeed, the coherence of the fertilizer utilization result with the preceding mechanism is what leads us to interpret the total farm expenditure divergence as a signal rather than instability in-



Notes: Top panel: DML residuals are out-of-fold nuisance residuals, $\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i)$; TWFE residuals are from an analogous specification estimated without the treatment terms, $\tilde{Y}_{it} = Y_{it} - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})$. Solid curves are third-degree polynomial fits. Bottom panel: absolute observation-level difference between the two residual series.

FIGURE 5

duced by rare treatment.

Table 5 presents treatment effects for the components of nonfarm expenditures. Receiving an agricultural loan increases food expenditures by 16.7 percent (0.155 log points) and non-food expenditures by 6.7 percent (0.065 log points), statistically significant at the 1 and 10 percent levels respectively. The interaction terms with farm size are statistically insignificant, consistent with the

Table 5.
Heterogeneous Effects of Agricultural Loans on Nonfarm Expenditures

	Food Expenditures (log)	Non-Food Expenditures (log)
Received Agricultural Loan (Loan)	0.155*** (0.060)	0.065* (0.035)
Loan $\times$ Farming land area (HA)	-0.055 (0.038)	0.029 (0.026)
<i>Specification</i>		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household CREs	Yes	Yes
<i>Learners</i>		
Loan $\sim$ X	PLH-L	PLH-L
Expenditures $\sim$ X	ERH-R	ERH-R
K-folds	5	5
<i>Diagnostics</i>		
R_Y^2 (Outcome: $Y \sim X$)	0.335	0.604
AUC (Loan $\sim$ X)	0.630	0.630
AP (Loan $\sim$ X)	0.082	0.082
Observations	5,898	5,900

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R_Y^2 reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

earlier finding that nonfarm expenditure responses do not vary systematically with landholdings. These results closely align with our TWFE specification in Table A7, consistent with our earlier assertions that agricultural credit broadly relaxes constraints on routine expenditures in a relatively mechanical fashion, with limited benefit from nonparametric estimation.

Although the estimated effects on food expenditures are sizable, they should not be interpreted as a one-to-one mapping from loan value to consumption. As shown in Table A3, the ratio of mean agricultural loan value to total nonfarm expenditures is relatively small (4.4–6.2 percent across quartiles). The estimated percentage increases therefore likely reflect broader household reallocation following liquidity relief, rather than direct spending of loan proceeds. In this sense, credit access appears to relax short-run budget constraints, allowing households to adjust consumption patterns alongside farm investment decisions.

4.4 Sensitivity to cross-fitting folds

We assess the stability of our core estimates to model specification. Specifically, DML results can be sensitive to the random partitioning of data during the cross-fitting process. Tables A8 and A9 demonstrate the sensitivity of our findings to model design by varying the number of cross-fitting folds ($K \in \{3, 5, 8\}$). The results indicate that our estimates are highly stable across fold specification. The greatest sensitivity is observed in Table A9, with a difference of approximately 0.005 between the 3-fold and 8-fold specifications. The majority of point estimates differ by less than 0.003 across fold specifications.

4.5 Robustness to Omitted Variable Bias

Approximately 76 percent of agricultural loans in our sample originated from informal or semi-formal sources (e.g. friends, money-lenders, savings associations). As a result, selection into credit is tied to opaque social factors. Consequently, unobservable confounders such as reputation, social networks, and entrepreneurial spirit represent a challenge to identification. This motivated our use of DML augmented for panel data, controlling for time-invariant unobservables with Mundlak means, and complex nonparametric relationships with our learners. The former design choice has been observed to significantly enhance performance over traditional linear TWFE specifications (Fuhr and Papies, 2024), while the latter has proven specifically useful in the context of noisy LMIC credit markets (Delavallade and Godlonton, 2023).

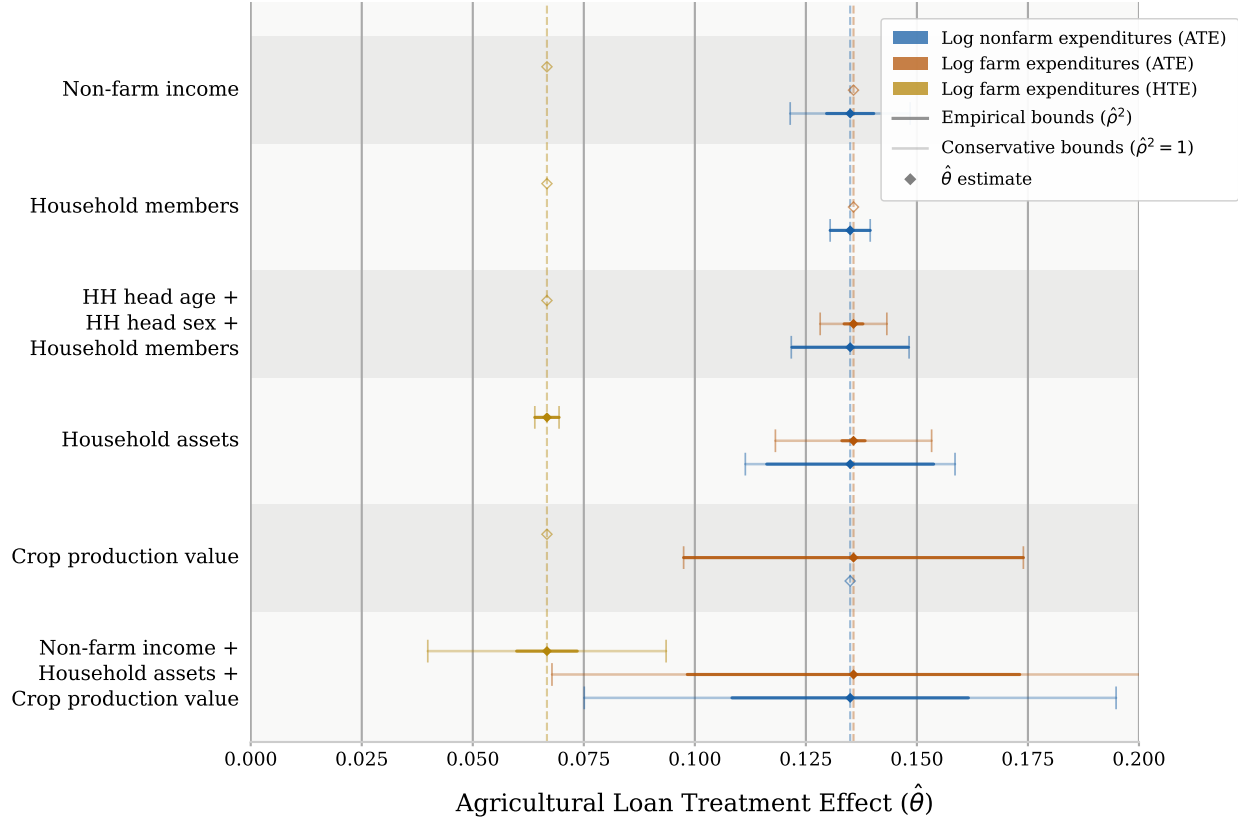
Nevertheless, to address the threat of time-varying omitted variable bias (OVB) we perform a sensitivity analysis of our results following Chernozhukov et al. (2026). Specifically, we benchmark our treatment effect estimates against pseudo-OVB estimated on observables. We do this by sequentially removing observables from our control vector X , measuring the impact on the predictive power of our nuisance functions, and estimating a maximally adversarial (i.e. perfectly correlated confounding between treatment and outcome) confidence bound.

Figure 6 illustrates the confidence bounds of our point estimates over the six most impactful omitted variable combinations. We focus our discussion around the benchmarked impact of removing our core endowment variables: nonfarm income, household assets, and crop production value. Removing these principal endowment variables shifts our treatment effect estimates toward zero by up to 44.4 percent for nonfarm expenditures, 50.0 percent for farm expenditures, and 40.2 percent for the farm expenditures interaction term.

This bounding exercise indicates that an unobserved confounder would need to be approximately 2 to 2.5 times as impactful on both credit access and expenditures as the combined effect of our core endowment variables to drive our point estimates to zero. More conservatively, an unobserved confounder would need to exert roughly the same magnitude of influence as our core endowment variables to render our nonfarm expenditure ATE point estimate statistically insignif-

Sensitivity of Treatment Effects to Omitted Variables

Estimated via Observed Variable Benchmarking bounds



Notes: Treatment effect bounds after omitting the indicated controls, following the observed-variable benchmarking approach of Chernozhukov et al. (2026). Empirical bounds use estimated confounding strength ($\hat{\rho}^2$); conservative bounds assume maximal confounding ($\hat{\rho}^2 = 1$).

FIGURE 6

icant at the 10 percent level, and about one-quarter the magnitude to do the same for the farm expenditure ATE point estimate.

While soft variables such as reputation play an important role in accessing credit in informal settings, the magnitude of time-varying confounding necessary to nullify our results appears implausibly large. Regarding entrepreneurial drive, household expenditures remain fundamentally bounded by available assets and liquidity. Because soft variables are unlikely to dictate aggregate spending to the degree of fundamental economic indicators, we conclude that our findings are robust to unobserved confounding.

5 Discussion and Limitations

This paper examines how agricultural credit shapes household expenditure allocation in Nigeria and how these responses vary with farm scale. The results indicate that credit raises both farm and nonfarm expenditures, and that the two responses differ in how they relate to farm scale. The farm investment response is concentrated among larger landholders and operates through fertilizer utilization, whereas the nonfarm expenditure response is broadly invariant to scale. These patterns align with models of production under incomplete markets, where the returns to purchased inputs depend on scale and complementary constraints (Carter and Olinto, 2003; Feder, 1985).

For relatively larger smallholders, agricultural credit is associated with stronger farm investment responses, identified through the fertilizer intensity channel. This is consistent with credit and landholdings acting as complements: where scale permits profitable intensification, credit can facilitate productivity-enhancing input use.

By comparison, the scale insensitive nonfarm routine expenditure response is itself informative. The effect on nonfarm expenditures is positive but does not present evidence of varying with farm size (the interaction coefficient is small, negative, and statistically indistinguishable from zero under both DML and TWFE specifications across the land distribution). Whereas the farm investment response is conditional on production capacity (proxied by farm size), the nonfarm consumption response is not, which points to credit relaxing a general household liquidity constraint rather than a farm specific one. The two channels are therefore distinguished by their relationship to scale: flat for consumption, increasing for farm investment. It is this divergence, more than the average treatment effect, that the expenditure allocation approach recovers.

The pattern also sharpens the interpretation of smaller farms. Smaller farms realize a similar consumption smoothing response to that of larger households but either face lower perceived returns to credit-induced intensification or more tightly binding liquidity constraints for consumption. As a result, smaller landholders proportionally invest less when receiving credit. This is consistent with the larger agricultural finance and microcredit literature showing that credit impacts are highly heterogeneous and shaped by underlying production opportunities and risk environments (Karlán et al., 2014; Hussain and Thapa, 2016; Meager, 2019).

From a food systems perspective, the results highlight the dual role of agricultural credit. For some households, credit supports productive investment and potential yield gains; for others, it primarily eases short-run liquidity constraints. Evaluating agricultural finance policies solely on average production responses may therefore obscure important welfare and resilience effects. Designing credit instruments that account for farm heterogeneity may improve alignment between policy objectives and household behavior.

Several limitations warrant consideration. First, although the DML framework controls for

high-dimensional observed confounding and incorporates correlated random effects to control for time-invariant unobserved confounding, identification relies on the assumption of conditional unconfoundedness. This is particularly relevant in settings where informal lending dominates and selection may depend on unobserved factors such as reputation or social networks. While the sensitivity analysis in Section 4.5 supports the robustness of our results to unobserved confounding, our identification strategy ultimately relies upon observables and cannot be held to the same causal certainty that a randomized design affords. Second, loan magnitudes are weakly correlated with land endowments in our sample, suggesting that credit constraints remain binding across the farm size distribution. Consequently, the results reflect a context of pervasive credit rationing and may not generalize to environments with deeper formal financial penetration.

An additional limitation concerns measurement of food expenditures. Our food expenditure data are constructed from annualized 7-day recall data, which may introduce aggregation error and inflate expenditure levels relative to observed income streams. While our analysis focuses on percentage changes rather than levels, any systematic measurement error in food reporting could affect the magnitude of estimated effects. We therefore interpret the food expenditure results as indicative of directional responses rather than precise level shifts.

Finally, the estimated percentage effects on food expenditures are large relative to mean loan size. This pattern suggests that credit access may induce broader expenditure reallocation rather than direct mechanical spending of loan proceeds. However, the data do not permit direct observation of intra-household allocation of loan funds, and we therefore interpret these responses as equilibrium adjustments following liquidity relaxation rather than as precise mappings from credit to specific expenditures.

6 Conclusions

This paper investigates how agricultural credit affects expenditure allocation among smallholder farmers in Nigeria and how these effects vary with farm scale. Using panel-adjusted Double Machine Learning, we estimate both average and heterogeneous treatment effects while accounting for high-dimensional confounding and time-invariant heterogeneity (Chernozhukov et al., 2018, 2025; Semenova and Chernozhukov, 2020).

Our findings indicate that agricultural credit raises both farm and nonfarm expenditures, and that the two responses differ in how they relate to landholdings. For relatively larger smallholders, credit is associated with stronger farm investment responses, identified through fertilizer intensification. The nonfarm routine expenditure response, by contrast, is positive but broadly invariant to farm size, pointing to relief of a general household liquidity constraint rather than a farm specific one. For smaller farms, expenditure adjustments are more concentrated in nonfarm categories. Specifically, smaller farms realize the common nonfarm consumption smoothing response, but,

whether because of lower perceived returns to scale or more tightly binding liquidity constraints, do not exhibit the scale conditional investment response. These results are consistent with models of agricultural production under incomplete markets, where the profitability of purchased inputs depends on scale, supervision, and complementary constraints (Carter and Olinto, 2003; Feder, 1985). They also align with evidence from the broader microcredit and agricultural finance literature showing that credit functions as part of an integrated household portfolio rather than as project-specific capital (Angelucci et al., 2015; Banerjee et al., 2015; Meager, 2019).

By documenting continuous heterogeneity across the farm size distribution, this study helps reconcile mixed findings in prior work. Experimental and quasi-experimental studies have found positive investment responses when complementary conditions—such as insurance or favorable timing—are present (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2020). In settings characterized by pervasive credit rationing and exposure to shocks, however, liquidity relief may primarily facilitate consumption smoothing or risk management. Our Nigerian case reflects this duality by showing that credit also acts as a complementary condition for agricultural investment but only for larger smallholders.

Several caveats remain. Identification relies on the assumption of conditional unconfoundedness given a rich control vector and correlated random effects structure. Although the DML framework mitigates overfitting and high-dimensional bias (Belloni et al., 2014; Chernozhukov et al., 2018), informal lending environments may involve unobserved selection factors such as reputation or social ties. In addition, loan magnitudes are only weakly correlated with land endowments in our data, suggesting that credit constraints bind across the farm size distribution. The results therefore characterize behavior under severe credit rationing and may not generalize to contexts with deeper formal financial markets.

More broadly, the picture that emerges is of a single, fungible loan that households deploy according to both their consumption needs and agricultural investment capacity. It may support productive intensification where scale permits, while simultaneously enhancing resilience and consumption stability among more constrained farms. Designing financial instruments that recognize this heterogeneity, rather than targeting uniform production responses, may be central to advancing effective and equitable food system policy in low-income settings.

6.1 Policy and Methodological Implications

The findings have several implications for the design and evaluation of agricultural credit programs. First, the heterogeneous expenditure responses suggest that uniform loan products are unlikely to generate homogeneous outcomes. Credit appears to support productive intensification primarily among relatively larger smallholders, while functioning as short-run liquidity support among smaller farms. Credit design that accounts for farm scale (through differentiated

loan sizes, repayment structures, or complementary input-linked arrangements) may better align financial instruments with production capacity and household constraints.

Second, the results are directly relevant to large-scale public lending initiatives in Nigeria, including NIRSAL and the Anchor Borrowers' Programme (ABP). These programs operate in environments characterized by severe credit rationing and heterogeneous farm structures. Our evidence suggests that scale-sensitive guarantees, repayment structures, and targeting mechanisms may improve program effectiveness. For smaller farms, credit may primarily enhance resilience and consumption stability; for relatively larger smallholders, it may facilitate input intensification and yield growth. Recognizing these differentiated channels may help refine eligibility criteria, loan terms, and complementary services within such programs.

Third, the findings highlight the importance of integrating credit with complementary institutions. In settings where risk exposure and input access constraints remain binding, liquidity alone may not translate into sustained productivity gains among smaller farms. Prior experimental evidence emphasizes the role of insurance, savings mechanisms, and seasonal timing in shaping investment responses (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2020). Aligning agricultural credit programs with risk-sharing mechanisms and input markets may therefore enhance their impact.

Methodologically, this study illustrates both the value and the limits of flexible estimation in our setting. Benchmarked against analogous TWFE specifications, DML converges with the linear model on routine nonfarm expenditures and diverges on the farm investment margin. On the farm investment margin, expenditure is nonlinear in scale, and the agricultural intensification signal is only recovered through nonparametric nuisance estimation. The value of flexibility is therefore localized to contexts where the response function exhibits a high degree of complexity. This is informative to researchers: DML presents an advantageous estimation strategy for recovering treatment effects from observational datasets, but its benefit beyond traditional econometric tools is context dependent and can be diagnosed with relatively simple residual diagnostics.

More broadly, the evidence underscores that agricultural credit serves multiple functions within rural economies. Evaluations focused solely on average production effects risk overlooking welfare and resilience gains among households for whom credit primarily relaxes liquidity constraints. Designing financial architectures that acknowledge this heterogeneity is central to advancing effective and sustainable food system policy.

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Appendix A

TABLE A1.
Balance Table

Variable	Did Not Receive Agricultural Loan (N $\approx$ 5,604) Mean (SE)	Received Agricultural Loan (N $\approx$ 296) Mean (SE)	Difference (1) - (2) Diff.
Farming Land Area (Hectares)	1.184 (0.019)	1.345 (0.085)	-0.161*
Value of Farm Production (1,000s ₦)	354.15 (7.61)	445.92 (36.78)	-91.768**
Assets (1,000s ₦)	270.21 (7.19)	228.16 (26.14)	42.048
Nonfarm Income (1,000s ₦)	330.09 (7.67)	377.97 (39.69)	-47.879
Livestock Holdings (#)	0.772 (0.024)	0.672 (0.084)	0.100
Has Formal Land Rights (binary 0/1)	0.137 (0.005)	0.193 (0.024)	-0.056**
Income Shock (#)	0.200 (0.007)	0.176 (0.026)	0.025
Production Shock (#)	0.962 (0.016)	1.078 (0.071)	-0.116
Price Shock (#)	1.069 (0.018)	1.108 (0.083)	-0.039
Head Marital Status (binary 0/1)	0.776 (0.006)	0.817 (0.023)	-0.041*
Head Age	51.745 (0.199)	48.990 (0.813)	2.755***
Head Sex (binary 1 $\equiv$ Male)	0.815 (0.005)	0.845 (0.021)	-0.029
Number of Members in the Household	5.940 (0.046)	6.409 (0.184)	-0.469**
Number of Adult Members in the Household	2.932 (0.020)	2.959 (0.083)	-0.027
Household Has Phone Access (binary 0/1)	0.931 (0.003)	0.943 (0.014)	-0.011
Household Has Internet Access (binary 0/1)	0.349 (0.006)	0.365 (0.028)	-0.016
Probability of Moderate Food Insecurity	0.456 (0.006)	0.442 (0.025)	0.013
Food Consumption Score	48.618 (0.234)	50.243 (1.087)	-1.626
Received Non-Agricultural Loan (binary 0/1)	0.083 (0.004)	0.078 (0.016)	0.006

Notes: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Includes all observations in baseline and heterogeneous Nonfarm Expenditure models (i.e. observations with non-missing farm size values).

Observation counts vary slightly due to missing data.

TABLE A2.
Description of Key Variables

Treatment Variable	Description
Received Agricultural Loan (Binary 0/1)	Indicator variable, true when the household received a loan with a stated purpose for farm expenditures.
Outcome Variables	
Input Expenditures (1,000s ₦) ^a	Seeds, fertilizer, pesticides, machinery /tools.
Land Expenditures (1,000s ₦) ^a	Land purchases and rent.
Labor Expenditures (1,000s ₦) ^a	Hired labor, rented livestock labor.
Livestock Expenditures (1,000s ₦) ^a	Livestock purchases, livestock upkeep costs.
Farm Expenditures (1,000s ₦)	Sum of: Input, Land, Labor, and Livestock Expenditures.
Food Expenditures (1,000s ₦) ^a	Imputed value of food consumption, the product of purchased food consumed and respondent recalled market price. Averaged between pre-planting and post-harvest surveys.
Non-Food Expenditures (1,000s ₦) ^a	Household expenditures on goods (durable and non-durable, e.g. cigarettes, appliances), and services (e.g. vehicle repairs, insurance). Excludes: nonfarm business, education, health, debt, rent, vehicles, ceremonies, and holidays.
Nonfarm Expenditures (1,000s ₦)	Sum of Food and Non-Food Expenditures: intended to represent average daily consumption while excluding lumpy irregular consumption events.
Select Control Variables	
Farming Land Area (Hectares) ^{a,b}	Size in hectares of land owned by household, include actively harvested, fallow fields, and pasture land.
Value of Farm Production (1,000s ₦) ^{a,b}	Value of all crop production produced by a household, constructed from production statistics multiplied by applicable crop prices by region.
Assets (1,000s ₦) ^{a,b}	Sum of the estimated value of durable goods owned by a household (excluding dwelling and agricultural plots).
Nonfarm Income (1,000s ₦) ^{a,b}	Income from nonfarm businesses, remittances, and all other sources.
Livestock Holdings (#) ^b	The Tropical Livestock Unit value for a household in the post-planting survey. Provides a standardized measure of livestock density including all recorded livestock such as cows, sheep, and chicken.
Has Formal Land Rights (Binary 0/1) ^b	Related to agricultural plots, the household has documentation of land rights (at least one plot).
Head Marital Status (Binary 0/1)	Indicates if the head of the household is currently married, all other classifications such as divorced, separated, or widowed are classed as zero.
Food Consumption Score (#)	The Food Consumption Score (FCS) is an index that measures the diversity and frequency of food groups consumed, which is then weighted according to the relative nutritional value of the consumed food groups (ranges 0 to 100).
Received Non-Agricultural Loan (Binary 0/1)	Indicator variable, true when the household received a loan with a stated purpose besides farm expenditures.

^a Winsorized to top 1%

^b Imputation performed by Evans School Policy Analysis and Research Group ([UW EPAR, 2025](#)).

TABLE A3.
Summary Statistics by Farm Size Quartile

	Mean Full Sample	Q1	Mean by Farm Size Quartiles			
			Q2	Q3	Q4	
Agricultural Loan Variables						
Received Agricultural Loan (Binary 0/1)	0.05	0.04	0.04	0.05	0.06	
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	102.85	96.75	110.20	82.34	108.77	
Outcome Variables						
Input Expenditures (1,000s ₦) ^a	33.20	14.55	30.58	39.37	55.25	
Land Expenditures (1,000s ₦) ^a	4.29	2.63	4.32	4.38	6.32	
Labor Expenditures (1,000s ₦) ^a	64.56	31.55	58.17	66.04	102.44	
Livestock Expenditures (1,000s ₦) ^a	32.31	19.07	36.16	35.20	34.80	
Farm Expenditures (1,000s ₦)	117.98	56.71	112.89	131.52	188.76	
Food Expenditures (1,000s ₦) ^a	1,538.36	1,660.10	1,569.41	1,484.12	1,351.75	
Non-Food Expenditures (1,000s ₦) ^a	427.67	445.44	424.95	402.94	414.76	
Nonfarm Expenditures (1,000s ₦)	1,966.03	2,105.53	1,994.36	1,887.06	1,766.51	
All Fertilizer Use (kg/ha)	0.75	0.86	1.18	0.74	0.39	
Ratios						
Agricultural Loan to Total Farm Expenditures	87.2%	170.6%	97.6%	62.6%	57.6%	
Agricultural Loan to Nonfarm Expenditures	5.2%	4.6%	5.5%	4.4%	6.2%	
Control Variables						
Farming Land Area (Hectares) ^{a,b}	1.18	0.11	0.43	1.03	3.16	
Value of Farm Production (1,000s ₦) ^{a,b}	355.75	88.01	250.43	405.45	679.10	
Assets (1,000s ₦) ^{a,b}	276.68	316.97	259.25	236.84	280.37	
Nonfarm Income (1,000s ₦) ^{a,b}	349.87	367.59	361.43	322.12	299.64	
Livestock Holdings (#) ^{a,b}	0.79	0.21	0.45	0.82	1.56	
Has Formal Land Rights (binary 0/1) ^{a,b}	0.14	0.11	0.12	0.14	0.19	
Income Shock (#)	0.20	0.20	0.18	0.21	0.21	
Production Shock (#)	0.95	1.00	0.99	0.97	0.89	
Price Shock (#)	1.07	1.33	1.14	1.04	0.78	
Head Marital Status (binary 0/1) ^c	0.77	0.60	0.78	0.83	0.89	
Head Age ^c	51.74	55.56	51.17	49.86	49.96	
Head Sex (binary 1 ≡ Male) ^c	0.81	0.65	0.83	0.86	0.92	
Number of Members in the Household	5.93	4.48	5.81	6.33	7.21	
Number of Adult Members in the Household	2.93	2.48	2.83	3.08	3.34	
Household Has Phone Access (binary 0/1)	0.93	0.92	0.93	0.93	0.95	
Household Has Internet Access (binary 0/1)	0.36	0.41	0.34	0.32	0.34	
Probability of Moderate Food Insecurity	0.46	0.65	0.47	0.39	0.31	
Food Consumption Score	49.02	48.01	48.20	48.92	50.14	
Received Non-Agricultural Loan (binary 0/1)	0.09	0.10	0.09	0.07	0.07	

^a Winsorized to top 1%.

^b Imputation performed by Evans School Policy Analysis and Research Group ([UW EPAR, 2025](#)).

^c Characteristics of the Head of Household, coded verbatim from survey responses.

Note: We conducted ANOVA *p*-value tests of equality across quartile means and found all variables statistically significant (at least at the 10% level) except Income Shocks.

TABLE A4.
Wave 4 Summary Statistics by Farm Size Quartile

	Mean Full Sample	Q1	Mean by Farm Size Quartiles			
			Q2	Q3	Q4	
Agricultural Loan Variables						
Received Agricultural Loan (Binary 0/1)	0.05	0.05	0.04	0.05	0.07	
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	76.72	107.41	86.62	53.44	67.89	
Outcome Variables						
Input Expenditures (1,000s ₦) ^a	26.40	10.29	19.24	32.89	48.97	
Land Expenditures (1,000s ₦) ^a	4.57	2.91	4.03	4.84	6.90	
Labor Expenditures (1,000s ₦) ^a	66.87	38.40	56.24	64.53	108.20	
Livestock Expenditures (1,000s ₦) ^a	19.61	11.01	17.53	18.99	26.21	
Farm Expenditures (1,000s ₦)	106.99	57.17	89.63	114.84	184.43	
Food Expenditures (1,000s ₦) ^a	1,338.82	1,389.48	1,318.80	1,361.76	1,252.38	
Non-Food Expenditures (1,000s ₦) ^a	287.87	295.75	264.42	267.50	305.17	
Nonfarm Expenditures (1,000s ₦)	1,626.70	1,685.23	1,583.22	1,629.26	1,557.55	
All Fertilizer Use (kg/ha)	0.87	1.04	1.26	0.93	0.45	
Ratios						
Agricultural Loan to Total Farm Expenditures	71.7%	187.9%	96.6%	46.5%	36.8%	
Agricultural Loan to Nonfarm Expenditures	4.7%	6.4%	5.5%	3.3%	4.4%	
Control Variables						
Farming Land Area (Hectares) ^{a,b}	1.40	0.15	0.54	1.29	3.62	
Value of Farm Production (1,000s ₦) ^{a,b}	338.65	79.93	205.46	366.27	702.95	
Assets (1,000s ₦) ^{a,b}	193.05	209.40	155.39	180.41	213.76	
Nonfarm Income (1,000s ₦) ^{a,b}	272.69	289.36	244.82	265.33	239.32	
Livestock Holdings (#) ^{a,b}	0.93	0.23	0.55	0.95	1.89	
Has Formal Land Rights (binary 0/1) ^{a,b}	0.17	0.15	0.16	0.17	0.20	
Income Shock (#)	0.25	0.32	0.22	0.24	0.22	
Production Shock (#)	0.38	0.41	0.36	0.39	0.39	
Price Shock (#)	0.19	0.23	0.19	0.22	0.15	
Head Marital Status (binary 0/1) ^c	0.78	0.62	0.77	0.86	0.91	
Head Age ^c	50.02	53.17	49.92	48.18	48.59	
Head Sex (binary 1 ≡ Male) ^c	0.83	0.66	0.83	0.89	0.95	
Number of Members in the Household	5.91	4.67	5.57	6.29	7.26	
Number of Adult Members in the Household	2.88	2.55	2.74	2.97	3.28	
Household Has Phone Access (binary 0/1)	0.94	0.94	0.93	0.95	0.95	
Household Has Internet Access (binary 0/1)	0.36	0.40	0.33	0.33	0.34	
Probability of Moderate Food Insecurity	0.33	0.55	0.31	0.25	0.21	
Food Consumption Score	51.43	51.31	50.25	51.22	52.23	
Received Non-Agricultural Loan (binary 0/1)	0.10	0.13	0.10	0.08	0.07	

^a Winsorized to top 1%.

^b Imputation performed by Evans School Policy Analysis and Research Group ([UW EPAR, 2025](#)).

^c Characteristics of the Head of Household, coded verbatim from survey responses.

TABLE A5.
Wave 5 Summary Statistics by Farm Size Quartile

	Mean Full Sample	Q1	Mean by Farm Size Quartiles			
			Q2	Q3	Q4	
Agricultural Loan Variables						
Received Agricultural Loan (Binary 0/1)	0.05	0.03	0.05	0.06	0.06	
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	129.83	110.04	77.15	141.63	150.27	
Outcome Variables						
Input Expenditures (1,000s ₦) ^a	40.01	15.76	35.95	49.25	67.06	
Land Expenditures (1,000s ₦) ^a	4.02	2.36	4.54	4.11	5.64	
Labor Expenditures (1,000s ₦) ^a	62.27	26.26	52.75	72.97	97.20	
Livestock Expenditures (1,000s ₦) ^a	47.79	28.12	53.80	49.35	54.82	
Farm Expenditures (1,000s ₦)	128.98	54.20	119.23	155.17	205.06	
Food Expenditures (1,000s ₦) ^a	1,737.89	1,849.36	1,843.66	1,705.43	1,404.82	
Non-Food Expenditures (1,000s ₦) ^a	567.46	558.70	540.48	563.52	577.78	
Nonfarm Expenditures (1,000s ₦)	2,305.36	2,408.06	2,384.14	2,268.95	1,982.60	
All Fertilizer Use (kg/ha)	0.63	0.79	0.99	0.60	0.28	
Ratios						
Agricultural Loan to Total Farm Expenditures	100.7%	203.0%	64.7%	91.3%	73.3%	
Agricultural Loan to Nonfarm Expenditures	5.6%	4.6%	3.2%	6.2%	7.6%	
Control Variables						
Farming Land Area (Hectares) ^{a,b}	0.97	0.09	0.35	0.82	2.62	
Value of Farm Production (1,000s ₦) ^{a,b}	372.75	76.53	265.03	463.71	686.60	
Assets (1,000s ₦) ^{a,b}	360.31	417.63	310.93	349.51	348.07	
Nonfarm Income (1,000s ₦) ^{a,b}	427.05	424.75	429.65	421.51	385.06	
Livestock Holdings (#) ^{a,b}	0.65	0.19	0.42	0.67	1.18	
Has Formal Land Rights (binary 0/1) ^{a,b}	0.12	0.09	0.09	0.13	0.17	
Income Shock (#)	0.15	0.12	0.13	0.17	0.19	
Production Shock (#)	1.52	1.43	1.52	1.54	1.68	
Price Shock (#)	1.94	2.13	1.94	1.92	1.79	
Head Marital Status (binary 0/1) ^c	0.76	0.58	0.80	0.81	0.86	
Head Age ^c	53.47	57.16	52.66	51.40	51.99	
Head Sex (binary 1 ≡ Male) ^c	0.79	0.63	0.83	0.84	0.88	
Number of Members in the Household	5.94	4.24	5.89	6.58	7.18	
Number of Adult Members in the Household	2.98	2.40	2.88	3.19	3.46	
Household Has Phone Access (binary 0/1)	0.92	0.92	0.91	0.93	0.94	
Household Has Internet Access (binary 0/1)	0.37	0.44	0.34	0.34	0.33	
Probability of Moderate Food Insecurity	0.58	0.72	0.59	0.53	0.49	
Food Consumption Score	46.61	45.84	46.39	46.79	46.54	
Received Non-Agricultural Loan (binary 0/1)	0.07	0.08	0.08	0.06	0.07	

^a Winsorized to top 1%.

^b Imputation performed by Evans School Policy Analysis and Research Group ([UW EPAR, 2025](#)).

^c Characteristics of the Head of Household, coded verbatim from survey responses.

TABLE A6.
Heterogeneous Effects of Agricultural Loans on Disaggregated Farm Expenditures (TWFE)

	Input Exp. (log) (1)	Land Exp. (log) (2)	Labor Exp. (log) (3)	Livestock Exp. (log) (4)	Fertilizer (log kg/ha) (5)
Received Agricultural Loan (Loan)	0.101 (0.134)	0.440* (0.255)	0.000 (0.109)	0.314 (0.276)	0.134 (0.166)
Loan $\times$ Farming land area (HA)	0.093 (0.090)	0.058 (0.217)	0.086 (0.077)	0.072 (0.118)	−0.006 (0.079)
<i>Specification</i>					
Controls	Yes	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes	Yes
Household FEs	Yes	Yes	Yes	Yes	Yes
Adjusted- R^2	0.400	0.338	0.215	0.211	0.421
Within Adjusted- R^2	0.061	0.012	0.054	0.047	0.112
Observations	3,670	352	4,010	1,176	1,676

Standard errors in parentheses. Two-way fixed effects (TWFE) with cluster-robust standard errors at the household level.

Loan is evaluated at mean farm size.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A7.
Heterogeneous Effects of Agricultural Loans on Nonfarm Expenditures (TWFE)

	Food Expenditures (log)	Non-Food Expenditures (log)
Received Agricultural Loan (Loan)	0.179** (0.085)	0.085 (0.052)
Loan $\times$ Farming land area (HA)	−0.046 (0.049)	0.012 (0.036)
<i>Specification</i>		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household FEs	Yes	Yes
Adjusted- R^2	0.350	0.603
Within Adjusted- R^2	0.069	0.126
Observations	5,898	5,900

Standard errors in parentheses. Two-way fixed effects (TWFE) with cluster-robust standard errors at the household level.

Loan is evaluated at mean farm size.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A8.
Baseline Sensitivity Analysis for Farm and Nonfarm Expenditures

	3 Folds	5 Folds	8 Folds
Panel A: Log Farm Expenditures			
Received Agricultural Loan	0.137* (0.072)	0.136* (0.071)	0.135* (0.071)
R_Y^2 (Outcome: $Y \sim X$)	0.346	0.351	0.352
AUC (Loan)	0.619	0.626	0.622
AP (Loan)	0.083	0.083	0.082
Observations	5,476	5,476	5,476
Panel B: Log Nonfarm Expenditures			
Received Agricultural Loan	0.137*** (0.049)	0.135*** (0.049)	0.132*** (0.048)
R_Y^2 (Outcome: $Y \sim X$)	0.417	0.420	0.422
AUC (Loan)	0.622	0.630	0.625
AP (Loan)	0.081	0.082	0.080
Observations	5,900	5,900	5,900
<i>Specification</i>			
Controls	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes
<i>Learners</i>			
Loan $\sim X$	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R_Y^2 reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A9.
Heterogeneous Sensitivity Analysis for Farm and Nonfarm Expenditures

	3 Folds	5 Folds	8 Folds
Panel A: Log Farm Expenditures			
Received Agricultural Loan	0.126* (0.073)	0.124* (0.072)	0.125* (0.072)
Loan $\times$ Farming Land Area (HA)	0.068 (0.046)	0.067 (0.045)	0.067 (0.045)
R_Y^2 (Outcome: $Y \sim X$)	0.346	0.351	0.352
AUC (Loan)	0.619	0.626	0.622
AP (Loan)	0.083	0.083	0.082
Observations	5,476	5,476	5,476
Panel B: Log Nonfarm Expenditures			
Received Agricultural Loan	0.142*** (0.049)	0.139*** (0.049)	0.137*** (0.048)
Loan $\times$ Farming Land Area (HA)	-0.033 (0.031)	-0.031 (0.031)	-0.031 (0.031)
R_Y^2 (Outcome: $Y \sim X$)	0.417	0.420	0.422
AUC (Loan)	0.622	0.630	0.625
AP (Loan)	0.081	0.082	0.080
Observations	5,900	5,900	5,900
<i>Specification</i>			
Controls	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes
<i>Learners</i>			
Loan $\sim X$	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R_Y^2 reports nuisance learner performance for the outcome model.

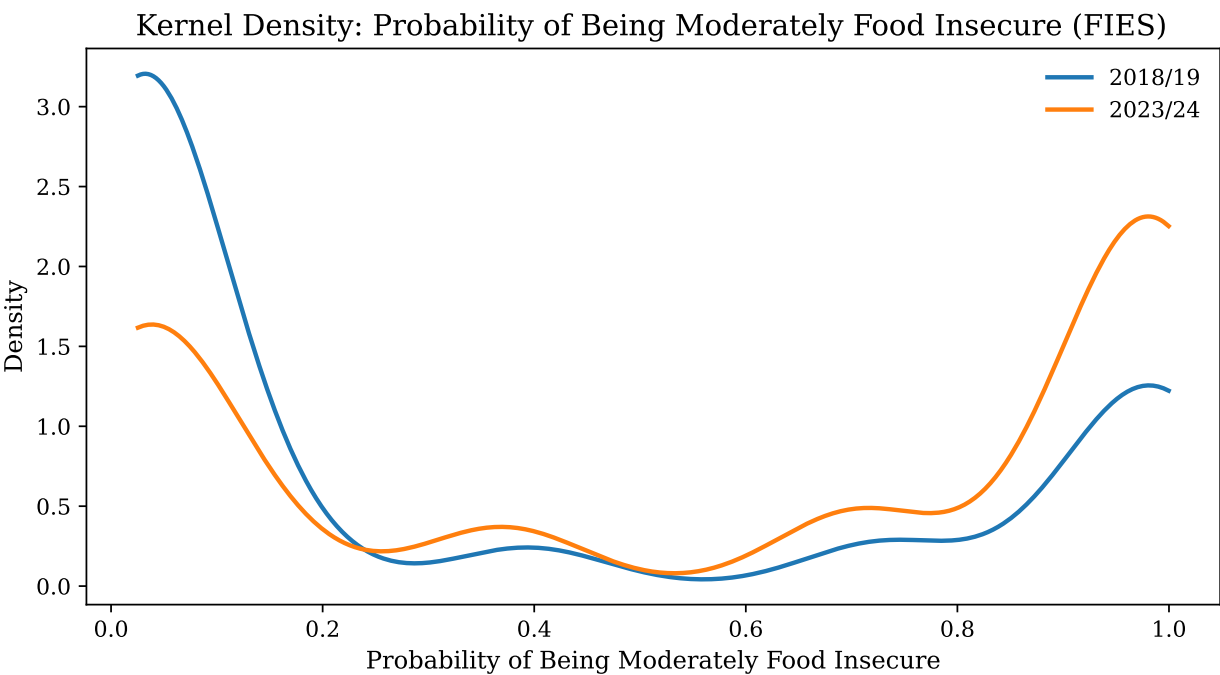
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

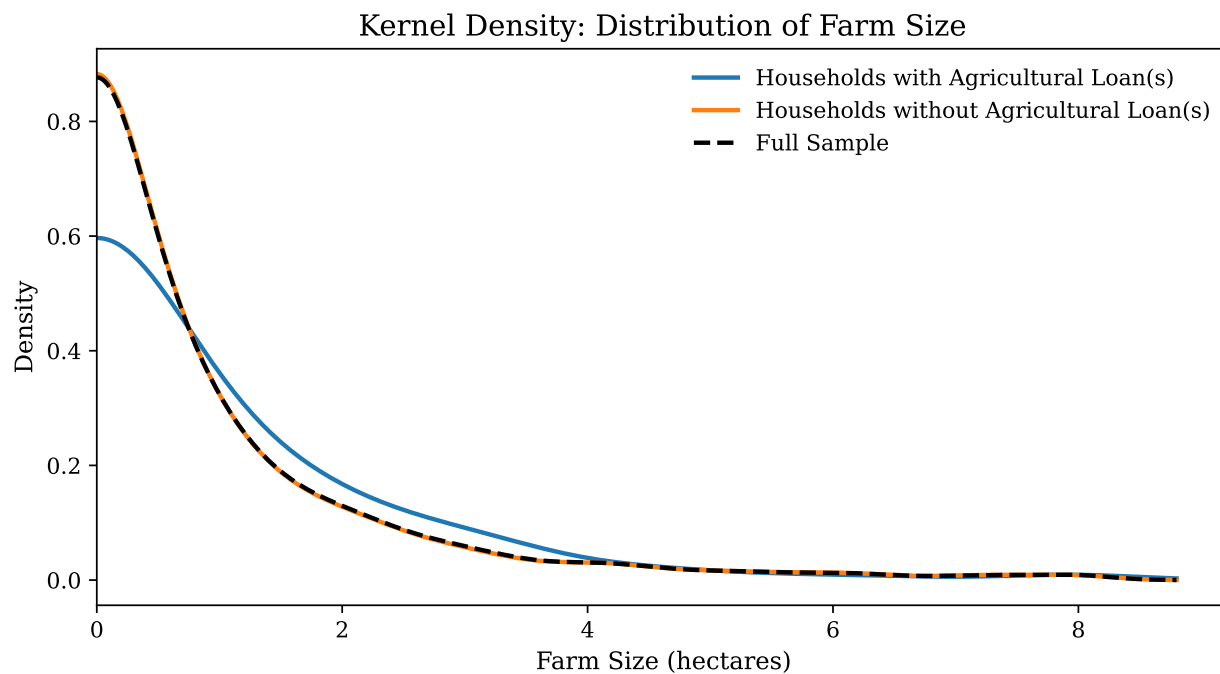
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Descriptive Figures



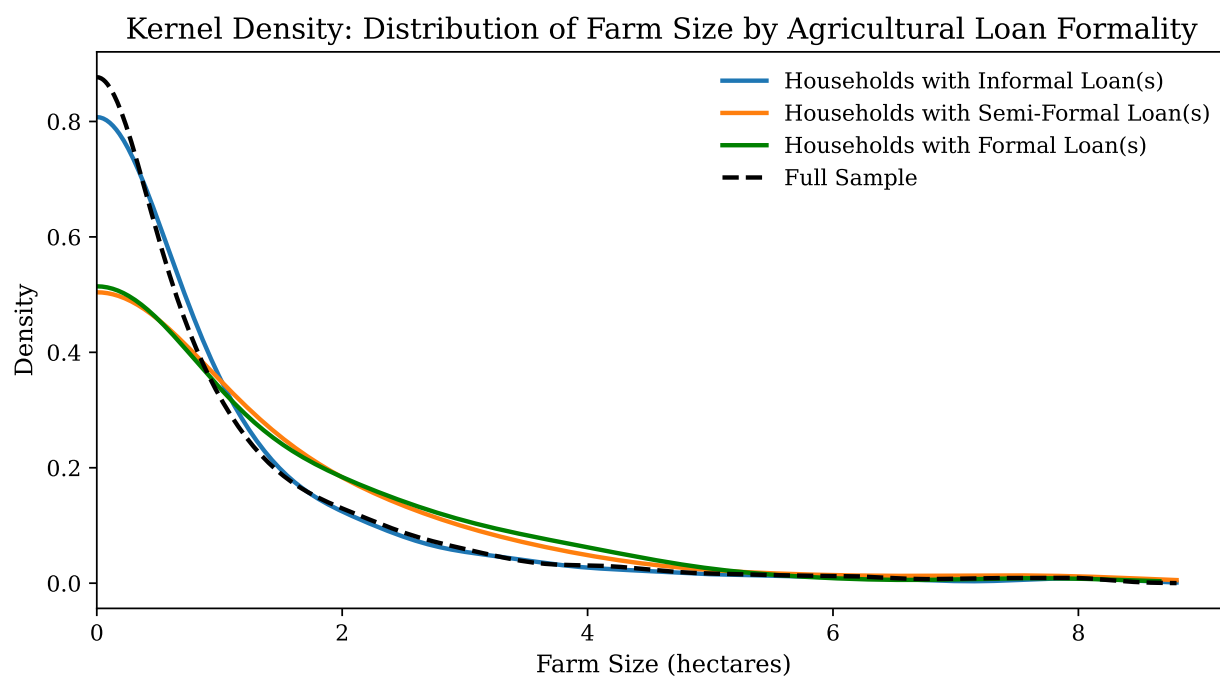
Notes: Kernel density of the probability of moderate food insecurity (FIES), by survey wave. Sample restricted to households observed in both waves.

FIGURE A1



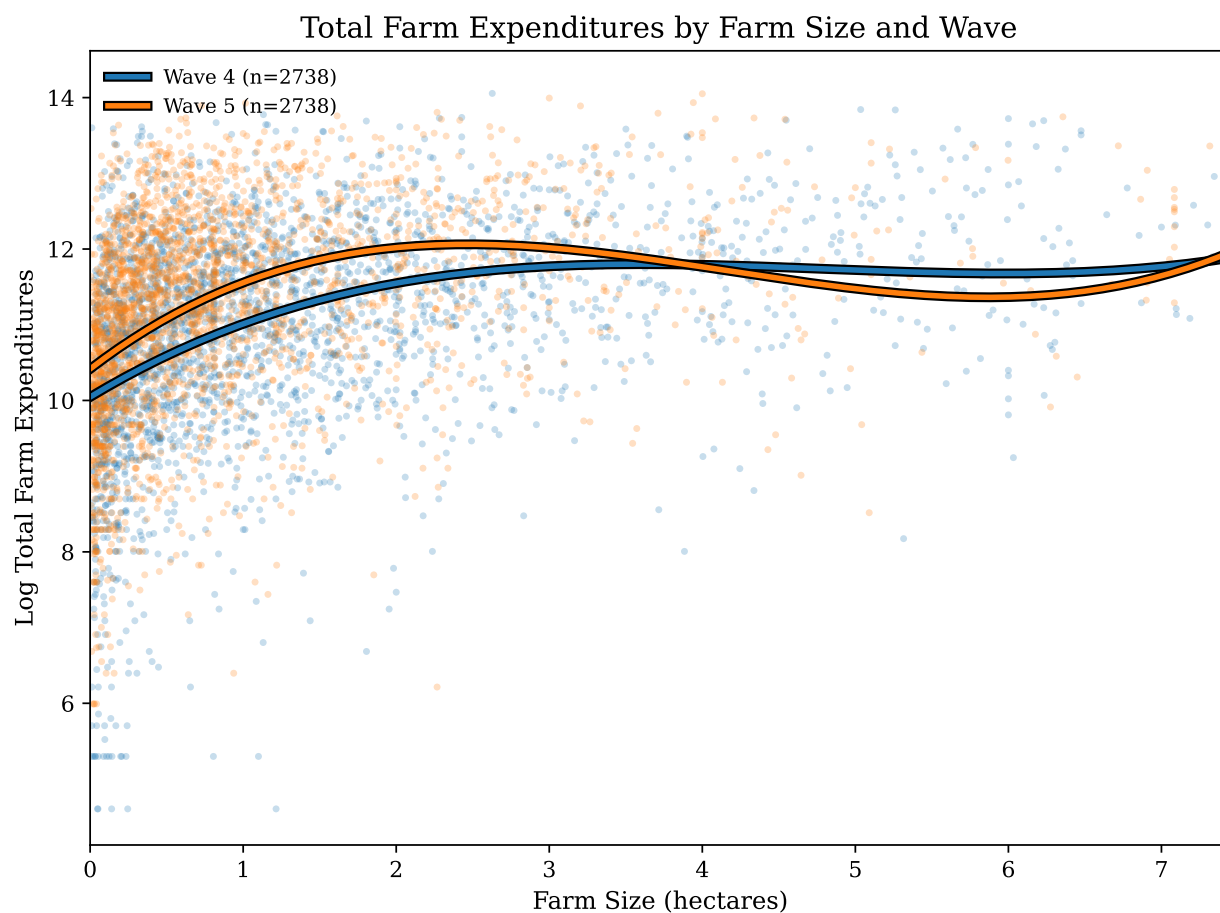
Notes: Kernel density of farm size by agricultural loan receipt.

FIGURE A2



Notes: Kernel density of farm size by agricultural loan formality.

FIGURE A3



Notes: Log total farm expenditures by farm size and survey wave, with third-degree polynomial fits.

FIGURE A4

Appendix B

The GATE estimates in Table B1 broadly align with the linear interaction reported in Table 3, though with less precision given the further dilution of the already rare treatment. In Panel B, we observe that the impact of agricultural credit on farm expenditures among the smallest landholders (Q1) is small and statistically insignificant (0.031) while the effect on the largest landholders (Q4) is large and significant at the 5 percent level (0.235). In contrast, Panel A shows a large effect of agricultural credit on nonfarm expenditures among the smallest landholders significant at the 10 percent level (0.207) with some attenuation among the largest landholders (0.112). The endpoints of the farm size distribution are therefore generally consistent with the pattern observed in our continuous specification, in which credit is highly fungible, relaxing consumption constraints among all smallholders while additionally supporting investment among larger smallholders.

The middle quartiles (Q2 and Q3), however, present results indicating a non-monotonic pattern that the continuous specification cannot capture. We attribute this partially to the loss of power: treatment effects are now identified with between 58 and 93 treated observations per quartile. Additionally, an unexpected pattern emerges between Panels A and B which is worth noting. Across quartiles, it appears treatment effects are inversely correlated, both in statistical significance and magnitude of point estimates. It should be noted that the farm and nonfarm expenditure variables are constructed independently, and as such, this result is not a mechanical artifact of the data cleaning process. Speculatively, the inverse relationship might point directly to the fungibility of agricultural credit, reinforcing the expenditure reallocation interpretation developed in Section 4. That said, with the loss of precision in the quartile analysis, these results are ultimately inconclusive but present an opportunity for future research.

Table B1.
Group Average Treatment Effects (GATEs)
by Farm Size Quartile for Farm and Nonfarm Expenditures

	Q1	Q2	Q3	Q4
Panel A: Log Nonfarm Expenditures				
Received Agricultural Loan	0.207* (0.106)	0.048 (0.086)	0.176* (0.094)	0.112 (0.100)
Treated	60	66	77	93
Observations	1,475	1,475	1,475	1,475
Panel B: Log Farm Expenditures				
Received Agricultural Loan	0.031 (0.167)	0.297** (0.149)	−0.019 (0.144)	0.235** (0.109)
Treated	58	59	77	87
Observations	1,369	1,369	1,369	1,369
<i>Specification</i>				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes	Yes
<i>Learners</i>				
Loan $\sim X$	PLH-L	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R	ERH-R
K-folds	5	5	5	5

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.